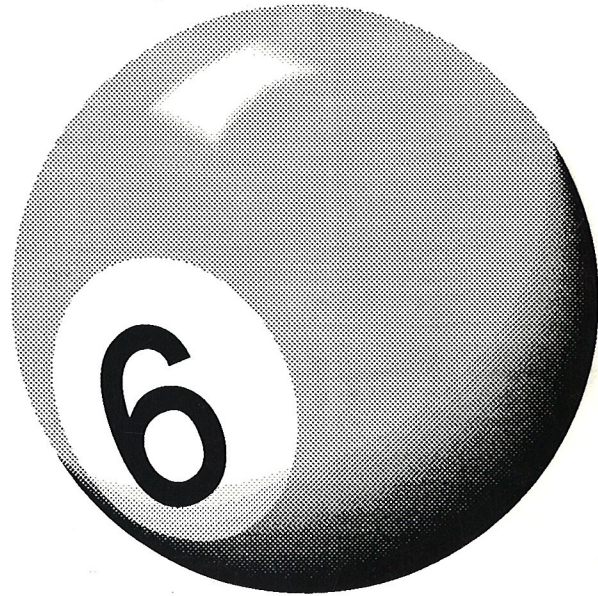
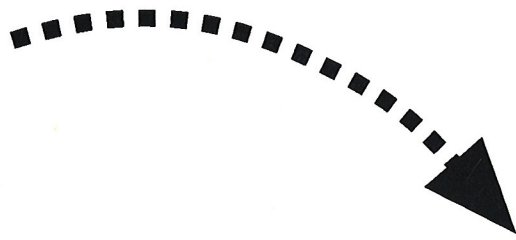


June 22/92

THE SYSTEM 6 BOOK



by Jerry Kindall



Quality Computers™

June 22/92

THE SYSTEM 6 BOOK

by Jerry Kindall

Exclusively for use with
Quality Computers' *Just System 6*
and *System 6 Bonus Pack* packages

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THE SYSTEM 6 BOOK by Jerry Kindall

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One Small Step...

Apple IIGS System Software 6.0 is a giant leap forward for the Apple II, the computer which launched the personal computer revolution. In its current incarnation as the Apple IIGS, the Apple II remembers the past with complete compatibility with older software, while looking into the future with a Macintosh-style graphic user interface and toolbox. Now with System 6, the IIGS gains even more speed, power, and compatibility. And the best part is that System 6 works on the computer you own now!

You've probably heard a lot about System 6, thanks to Apple. Developers saw a very rough version of System 6 at the A2-Central Summer Conference in July, 1991 and were mailed a pre-release version on CD-ROM later that year. Unexpectedly, these developers were actually encouraged to discuss System 6 on electronic information services and in other media. Apple software engineers, too, were permitted to talk about System 6 publicly, leading to an unprecedented amount of pre-release discussion.

We're pleased to tell you that most of what you've heard is true. System 6 will catapult your IIGS into the nineties and beyond. We think you'll enjoy the trip.

What's New

Here are the most important new features of System Software 6.0.

- *StyleWriter printer support.* Now you can use Apple's low-cost inkjet printer with any Apple IIGS Desktop program. System 6 also has more large fonts to take advantage of the capabilities of this and other high-resolution printers.
- *Macintosh, Apple DOS 3.3, and Apple Pascal disk support.* Now your IIGS can read Macintosh, DOS, and Pascal disks as easily as standard ProDOS disks. System 6 can format and write to Macintosh disks as well (DOS and Pascal disks can only be read). The IIGS still can't run Macintosh software, but exchanging data is easier than ever.
- *A completely redesigned Finder.* Faster, friendlier, and more powerful than ever before, with advanced features like tunneling and Finder extensions.

- *All-new desk accessories* featuring a Calculator and Find File. Even the Control Panel has been revamped—each Control Panel now appears in its own independent window, just like on Macintosh System 7.
- *A general rounding-off of rough edges.* The Finder can be set to avoid grinding your 5.25" drives. When the computer asks you to insert a disk it needs, you no longer have to hit Return—the computer detects it automatically. The screen no longer switches to text mode and back to graphics when launching some Desktop programs.
- *New music tools and applications* to allow new programs to sound even better. You may have heard of SynthLab, Apple's prototype music program—it's included.
- *Media-control toolset* and desk accessories to allow even easier integration of video with your multimedia presentations.
- *Special Aids* (Video Keyboard, CloseView, and Easy Access) for physically handicapped users.
- *More tools for programmers to write great programs.* Like drivers for Apple flat-bed scanners and SCSI tape drives. You may never use these tools directly, but they'll make writing quality IIGS software even easier. And eventually more and more IIGS programs will take advantage of special System 6 features.
- *Simpler and faster hard drive installation.* The new Installer's Easy Update option lets you install System 6 painlessly from the self-booting Install disk. If you need to customize your installation, of course, you can do that too!
- *ProDOS 8 Enhancements.* ProDOS 8 now remaps other hard drive partitions to unused slots, starting with Slot 1. At last—if you have a hard drive attached to an Apple SCSI card, you can now access *all* your partitions from ProDOS!

There's much more, of course. System 6 is a project that took a team of skilled programmers over a year to complete. To learn more about the new features of System 6, look for the sections throughout this book labeled with a ®.

System Requirements

System 6 requires a minimum of 1.25 megabytes of RAM with ROM 01 (the standard configuration plus a 1 megabyte expansion card), or 1.125 megabytes of RAM on ROM 03 (the standard configuration), and a single 3.5" drive. A hard disk drive is highly recommended (hard drive installation requires 2 megabytes of RAM). Additional memory and/or an additional 3.5" drive are highly recommended. System 6 also supports many other peripherals, including printers, modems, 5.25" drives, and modems.

Contents

We start out by telling you how to get started with System 6—by making backup copies and installing System 6 on your hard drive, if you have one. Next, we talk about how System 6 and the Apple IIGS Desktop work—this is must-read information if you're not familiar with using Desktop programs. Then we talk about the Finder and System 6's desk accessories, control panels and utilities.

Each section is divided into chapters, which are further divided into logical divisions that make finding the material you need simple. We hope you find this book an enjoyable and useful reference as you use System 6.

Conventions

We've made this book easier to navigate by differentiating the chapter titles and subheadings in different sizes and styles of Optima, an extremely easy-to-read typeface. We also draw your attention to the types of information below with special formatting:



Note!

These paragraphs contain essential information that requires your complete attention. Understanding this information is very important to effective use of System 6.



Warning!

These paragraphs alert you that you might damage equipment or lose data if you don't follow the instructions carefully. This icon is a sort of emphatic Note! icon.



Shortcut

These paragraphs alert you to time-saving techniques that can make your life easier.



By The Way...

These paragraphs contain interesting but nonessential information related to the topic at hand.



New & Improved

These paragraphs tell you how System 6 differs from previous system versions, pointing out completely new features or familiar features that work differently in System 6.

What Should I Read?

If you're new to the IIGS or have been using the IIGS only to run programs like AppleWorks and have little experience with the IIGS way of doing things, we suggest starting reading this entire book. If you're a seasoned IIGS user updating from System 5, you'll want to skim the Desktop and Finder chapters (looking for **⑥ New & Improved** sections and similar information) and pay close attention to the Utilities chapter.

About The Disks

The Apple Disks

Apple ships IIGS System Software 6.0 on six 3.5" disks. These disks are:

- *System Disk*—System 6 on a bootable stand-alone 3.5" disk
- *System Tools 1 & 2*—Additional system elements for installation
- *Fonts*—Additional fonts for users with high-resolution printers
- *Install*—A bootable 3.5" disk for performing System 6 installation
- *SynthLab*—The SynthLab application and sample music files

The Bonus Pack Disks

If you bought the System 6 Bonus Pack, you got six additional 3.5" disks:

- *Bonus Stuff*—Desk accessories, icons, clip art, ShrinkIt, FlashBoot, and HyperSound
- *Bonus Fonts 1 & 2*—More unique and interesting fonts to supplement Apple's selection
- *Bonus Sounds 1 & 2*—Sounds to spice up your IIGS desktop and hypermedia stacks
- *ZZCopy*—Utility for backing up 3.5" disks, plus shareware programs

Read This Before Doing Anything!

If You Don't Have A Hard Drive

If you don't have a hard drive, we heartily suggest making the investment. While System 6 will run fine on 3.5-only systems, you may end up with longer waits and, if you have only a single 3.5 drive, more disk swapping than with previous versions of the IIGS System Software. A good, fast hard drive is the single most significant improvement you can make to your computer because it allows you to store not only your System Software but all your programs and data on a single high-speed storage device. Don't take our word for it—ask anyone who already uses a hard drive on their IIGS. If you don't have the funds for a hard drive, an additional 3.5 drive or more memory can provide many of the same benefits at a lower cost.

If you decide not to get a hard drive, you won't really “install” System 6. You'll just start up the computer from the System 6 System Disk, just as you started up the System 5 (or earlier) System Disk when you wanted to use that System version. You may find that the additional features of System 6 are worth the occasional disk swap and choose to run under System 6 all the time, or you may decide to use System 6 only when you need its special features and start up directly into other programs via whatever System Software they came with. It's your choice.

If you have more than two megabytes of memory installed in your Apple IIGS, you may want to reserve some of your memory for use as a RAM Disk and install System 6 on the RAM Disk. See the FlashBoot manual for more information on using your IIGS RAM Disk and our exclusive FlashBoot utility.

Note!

Be sure to back up your System 6 disks before using them. See the section after the next one for instructions on using the ZZCopy program.

The System Disk included with the System 6 package is a *minimum* configuration. It's adequate for running most programs, but a few programs may require additional tools, fonts, or drivers to be installed. The System Disk has a little free space on it, so you can make another backup of the original System Disk, install the additional software on it, and use that as a special-purpose System 6 startup disk for a particular program. For more details on the Installer, see the hard drive installation section in this chapter.

If You Do Have A Hard Drive

We suggest resisting the temptation to install everything at once. That makes getting used to System 6 more difficult than it needs to be, and it also makes diagnosing your problems harder too. First install the System 6 "Easy Update" (hard drive installation instructions will follow in a moment). Then browse through this book and learn about System 6's new features. Skip over any sections that talk about software you haven't installed yet. Then install any additional System 6 features you want, and get used to them, too. Finally, install the System 6 Bonus Pack if you purchased it, and learn to use its features.

Remember, even seasoned users can feel overwhelmed by adding half a dozen disks' worth of new software to their computing environment.

Note!

Before doing anything, read this entire chapter and make sure you understand how to install System 6. If you don't, call our Technical Support department for clarification.

Making Backup Copies

Before beginning installation of System 6, you should make a set of backup disks. (The same is true of any program you get. Back up the originals, put the originals in a safe place, and use only the backups.) Yeah, we know some of you won't make the backups because there are so many disks—that's not our fault, though. **MAKE THE BACKUPS!**

By The Way...

If you have to call our technical support line to get a new copy of a disk you somehow messed up, the first thing our tech guys are going to ask is, "What happened to your backups?" and you're going to feel really, really stupid. Of course, if the disk was bad right out of the box we'll replace it promptly and we won't laugh at you at all.

If you have a favorite utility you use for making backup copies of disks (*Copy II Plus*, for example), feel free to use it to back up your disks and skip this section. The *FastCopy* program that comes with AppleWorks is quite simple to use and will do an admirable job of backing up the System 6 disks. Use whatever you're most familiar with.

If you bought the System 6 Bonus Pack, we've also included a utility called ZZCopy which can be used to make backup copies of the System 6 disks. To use this program to make backup copies of your disks, follow these steps:

1. Make sure the IIGS Startup Slot is set to "Slot 5" using the Control Panel. (If you're not familiar with the Control Panel, consult your Apple IIGS Owner's Manual.)
2. Put the ZZCopy disk in your first (or only) 3.5" drive and start up the computer (turn on the computer if it's off, or hold down ⌘ and Control while pressing Reset if the computer is already on).

3. When ZZCopy is finished loading, you'll see a picture of the designers of ZZCopy and your IIGS will say "Welcome." Then the ZZ Copy disk will be ejected and message "Insert Source" will appear.
4. Grab one of your original System 6 disks (it doesn't matter which one you start with as long as you copy them all) and open the write-protect slide, then insert it in your first 3.5" drive. ZZCopy will read the disk, then display the message "Insert Target."
5. Insert a blank or reusable disk. (If you have two disk drives, put it in your *second* drive.) *This disk will be completely erased, and ZZCopy won't ask you if it's OK to erase it first, so make sure there's nothing important on it!* ZZCopy will write a copy of the original disk onto your backup, then display the message "Insert Source."
6. Repeat steps 4 and 5 until you've backed up all the System 6 (and Bonus Pack) disks, including the ZZCopy disk.
7. Don't forget to set the Startup Slot back to your usual setting—normally "Scan"—using the Control Panel.

If, during the copy operation, ZZCopy detects an error, the IIGS will say "Alert!" and display an error message on the screen. If the error is on your target disk (the one you're copying onto), throw out the bad disk and go back to Step 4 to reload the original disk. Use a different target disk this time. If there's an error on the source disk (an original), call Quality Computers and we'll send you a new copy of that disk.

Installing System 6 On Your Hard Drive

Keeping Existing System Software

If your hard drive already has System 5 (or another older system) installed, you may want to keep it around until you're used to System 6 and are sure it's compatible with all the programs you're already using. (Plus you'll be able to move your favorite desk accessories, fonts, and other system enhancements from the old System to the new System and back.)

The easiest way to keep System 5 on your hard drive while installing System 6 on the same drive is to *rename the System and Icons folder*. You don't *have* to do this, and we definitely suggest you don't do it just yet—read this whole section before doing anything—and make sure you've made backup copies of your System 6 disks first.

Here's how (we'll assume you're already familiar with using your hard drive, at least a little):

- Start up your system as you normally do, booting from the hard drive.
- If any windows are open, close them all by pulling down the "File" menu and selecting "Close All."

- Double-click your startup disk's icon (in the upper right corner of the screen) to open its window.
- Make sure there's at least 1 megabyte (MB) free by looking at the information bar in the window you just opened. If there's not at least that much disk space free, you should free up some space (by deleting unused files or moving information to other hard drives or disks) before trying to install System 6.
- Scroll the window until you find the System folder, a folder with a rainbow apple on it.
- Click the System folder once, then type "System5" (without the quotes) on the keyboard, and press Return.
- Find the Icons folder in the window. Click it once and type "Icons5" (again omitting the quotes) and press Return.

Warning!

Once you've performed this procedure, you won't be able to start up from your hard drive until you've installed new system software on it. You should install System 6 immediately after performing this procedure.

Preparing For Installation

Make sure your IIGS's Startup Slot is set to Slot 5, so that the computer starts up from the 3.5" floppy drive. (You can do this using the IIGS Control Panel. If you're not familiar with the Control Panel, see your Apple IIGS Owner's Manual.)

System 6 includes a self-booting installation disk that includes the driver for the Apple SCSI Card. This means that System 6 will recognize most hard drives automatically. If you have a SCSI-compatible hard drive connected to your computer via an Apple SCSI Card, start up your system from the Install disk and proceed directly to the installation instructions.

If you have an Applied Engineering Vulcan hard drive, you must boot from a disk containing the Vulcan driver before you can run the installer. If you don't, the Installer will think you have ejected your hard drive during the installation and ask you to "Insert the disk AE1."

There's not enough room on the Install disk for the Vulcan driver. The System Disk does have some free space on it, so we suggest installing the Vulcan driver on the System Disk (placing it in the Drivers folder, inside the System folder, as described in the Vulcan manual), starting up from that disk, inserting the Install disk after the Finder appears, and double-clicking the Installer to start it up. If you have a single 3.5" drive, you'll be in for some additional disk swapping, but installation isn't something you do everyday, anyway.

If you have a RamFAST/SCSI hard drive controller, you don't need its driver installed to perform System 6 installation. The Install disk will work fine for installing System 6 on a hard drive connected to a RamFAST/SCSI card; just start up from the Install disk and proceed.

Easy Update

The Installer's "Easy Update" screen (Figure 2A) is fairly straightforward. There's a message that tells you what disk System 6 will be installed on—in our example, it says "Click 'Easy Update' to install Apple II GS System Software on the disk 'Q1'." If the name of the disk is wrong, use the mouse to move the arrow to the "Change Disk" button, like it is in our example. Click the mouse button once and wait to see what disk appears next. (It may take a second or two for the next disk's name to appear). If that's not the right disk, click the mouse button again and wait for the next disk name to appear—keep doing that until the name of the disk you want to install System 6 on appears on the screen.

▲ By The Way...

If you have a Q Drive, you probably want to install System 6 on the disk named "Q1." If you have a Vulcan, you'll probably want to install to "AE1." If you have some other hard drive, the name will probably be different, but you should recognize it when it appears. In any case, you want to install System 6 on the first partition on your hard drive.

⚠ Note!

If the correct disk name never appears, contact Technical Support for help. If, on the other hand, it appears briefly and then disappears, you're clicking the "Change Disk" button too quickly. Remember to wait until the new disk name appears on the screen before clicking "Change Disk" again.

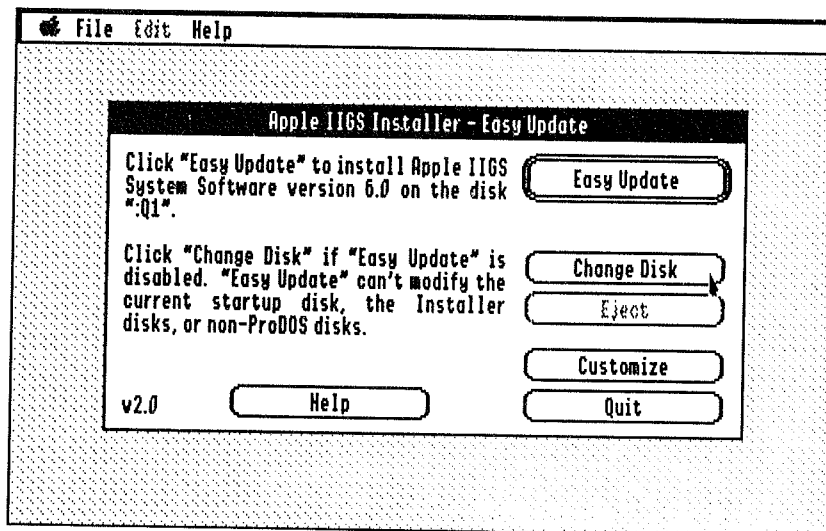


Figure 2A—System 6 Installer: Easy Update Screen

Once you've selected the disk you want to install System 6 on, move the arrow to the "Easy Update" button and click the mouse button. Installation will begin. You'll be prompted to switch disks a few times—just remove the disk the computer is finished with, and insert the next disk. The IIGS will automatically detect that you've inserted the new disk and continue—there's no need to press Return or click the on-screen "Ok" button.

 Note!

If the Easy Update button is "dimmed" (i.e., the words "Easy Update" are displayed in gray instead of in black), you're probably trying to install System 6 on the disk you started up from. Start up from the Installer disk (or from the System disk containing the Vulcan driver, if you have a Vulcan hard drive) and try again.

After installation is complete, set the Startup Slot back to its usual value ("Scan" or "Slot 7" in most systems) using the Control Panel. Hold down ⌘ and Control while you press and release Reset to restart the computer and enter the world of System 6.

If you have an AE Vulcan hard drive, an AE High Density 3.5" drive, a RamFAST/SCSI card, or any other third-party device requiring a driver (such as Hewlett-Packard printers) install their drivers according to the directions that came with the device. (If the driver software is old, it might be a good idea to contact the manufacturer to make sure the driver is compatible with System 6 before installing it.)

You *can* install additional system software features right away, as well, but we suggest you get used to System 6's basic features before you add any other features, especially if you're fairly new to the IIGS. Read through this manual (particularly the chapter on the Finder) and then, when you're comfortable with System 6, install additional features like Macintosh disk compatibility, networking, sounds, and drivers for your other peripherals like CD-ROM.

Installing Additional Features

As we mentioned in the last section, the Installer can install many System 6 updates, not just the "Easy Update." Once you're familiar with the basic System 6 features, you can install additional features as easily as you installed the original system software.

Remember, each additional feature you install needs more memory and more disk space. If you plan to use all of System 6's features, we suggest a minimum of 2 megabytes of RAM—and the more, the better.

To install additional System 6 features, get into the Installer as described in the section above and choose a disk using the "Change Disk" button. Then click the "Customize" button. A list of available system updates (Figure 2B) will appear. Use the scroll bar to the right of the list to look at the entire list of updates; select the updates you want to install from the list.

To select more than one update to be installed at once, hold down the ⌘ key while selecting the second and subsequent updates. (The same maneuver can also be used to unselect an update that you selected accidentally.)

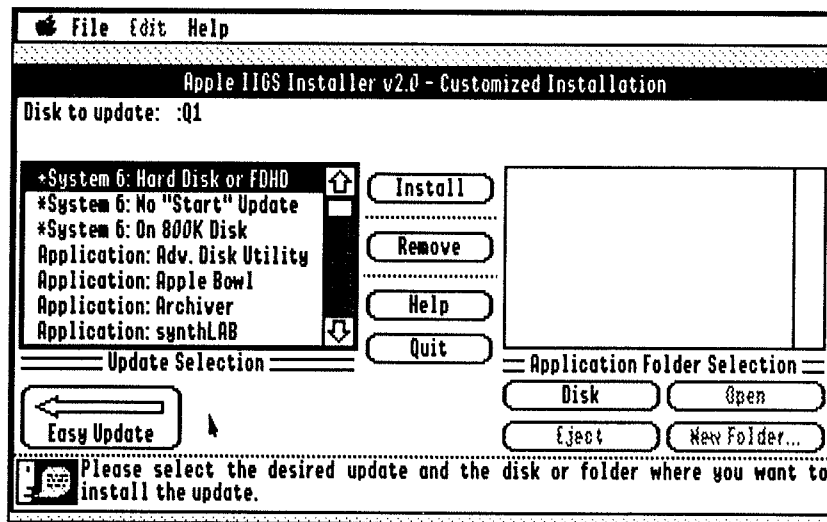


Figure 2B—System 6 Installer: Custom Installation

- *System 6: Hard Disk or FDHD*: Installs the “Easy Update,” plus some additional system software such as the Calculator and Find File NDAs, the HFS FST (for accessing Macintosh disks), the Sounds control panel, and the Special Aids software. This update requires at least a megabyte of disk space (and thus, you must install it on a hard drive or high-density 3.5” disk). We suggest that you install the “Easy Update” instead, and then install the other items that interest you, because the Special Aids are incompatible with several popular programs including AppleWorks GS.
- *System 6: No “Start” Update*: Installs the same software as above, but doesn’t update your existing Start application. (Use this update to completely replace older System Software with System 6 while keeping your current program selector, such as ProSel or Wings.)
- *System 6: On 800K Disk*: Installs the bare minimum System Software on a blank 3.5” disk or RAM Disk in less than 800K.
- *Application*: Installs the named application—Advanced Disk Utilities (useful for partitioning and formatting a hard drive), Apple Bowl (a bowling game), Archiver (a backup utility for 3.5” and hard drive users), SynthLab (a new music program), or Teach (a simple word processor). We suggest Archiver and Teach as definite must-install programs, and SynthLab is great if you’re at all interested in music. Having the Advanced Disk Utilities on your hard drive doesn’t seem that useful to us—if you format or partition the drive, everything, including ADU, will be erased.
- *Control Panel: Sound*: Installs the Sound control panel and a few sample sounds. This is a definite must-install for most hard drive users—it’s useful, and a lot of fun, too.

- *Desk Acc:* Installs the named desk accessory, the Calculator or Find File. Find File is a must-have for hard drive users; the Calculator is also useful, but there's a better one in the System 6 Bonus Pack.
- *Drive and Driver:* Installs drivers for the named devices. The "Easy Update" automatically installs Apple Disk 5.25", Apple Disk 3.5, and SCSI Hard Disk drivers if you have those devices attached to your computer. If you have other devices, install them here.
- *File System:* Installs File System Translators (FSTs) for Macintosh (HFS) disks, Apple DOS 3.3 disks, and Apple Pascal disks. The HFS FST is highly recommended; it allows you to read, format, and write Macintosh disks in your Apple IIGS drives, and additionally lets you access Macintosh hard drives attached to your computer. The DOS 3.3 and Pascal FSTs allow you to read (not write) 5.25" disks in these older disk formats.
- *Fonts:* The "Easy Update" option installs "Standard" fonts if there's enough room or "Minimum" fonts otherwise. We recommend that hard drive users install "All" fonts.
- *Media Control:* These install drivers for various types of video and audio disc players, along with the Media Control Toolkit, which allows you to control these devices as part of multimedia presentations. *Media* drivers should be installed if you have a MIDI synthesizer interface; VideoMix should be installed if you have a Video Overlay Card.
- *Network:* Network updates should be installed if you're attached to an AppleTalk network and want to share files or print to a shared printer.
- *Printer:* Install the correct update(s) for your printer(s), including network (marked as "ATalk") printers such as the LaserWriter, ImageWriter, and ImageWriter LQ.
- *Special Aids:* These are tools for helping physically handicapped users use the IIGS. CloseView magnifies the screen; Easy Access provides "Sticky Keys" and "Keyboard Mouse" features; Video Keyboard provides an on-screen keyboard that users can click instead of typing on the keyboard. We do *not* recommend Special Aids for most users, because some are incompatible with popular programs such as AppleWorks GS.

Once you've selected the update(s) you want to install, click the "Install" button to begin installation. The Installer will prompt you to swap disks when needed; just eject the old disk and insert the correct disk. The Installer will automatically detect the new disk and continue the installation.

When the installation is complete, change the Startup Slot (in the Control Panel) back to its usual setting—usually "Scan"—and restart the system to begin using the updates.

Summing up, then, we recommend the following updates for most hard drive users:

- Easy Update to start with
- Archiver, Teach, and SynthLab applications
- The Sound control panel
- The Find File and Calculator desk accessories
- Drivers for any disk drives you have attached to your IIGS
- All file System Translators (what the heck, you might as well)
- All fonts
- Driver(s) for your printer(s)

Warning!

When installing Apple system software, always use the Installer. The Installer will always make sure you have all the files you need for a particular installation. If you attempt to install system software manually, using the Finder or a similar tool, it's all too easy to miss files, or to not update files that need updating. This can cause unexplained system problems.

Installing The Bonus Pack

We recommend that you install the Bonus Pack only after you're thoroughly familiar with the features of System 6. Installing before you've read this manual will only confuse you, because some of the things we discuss elsewhere in this book apply only to standard Apple-issue System 6. The Bonus Pack adds a number of new features to System 6, but it is *not* provided by Apple.

The Bonus Pack is installed via a program called ShrinkIt. ShrinkIt allowed us to pack *lots* of extras into the Bonus Pack—much more “stuff” than would normally fit onto a disk—because it can actually make files smaller. ShrinkIt isn't quite as easy to use as the Installer, though, because it's a general-purpose program. We assume some familiarity with using Desktop programs (which you'll gain if you read this manual, or which you may already have).

To get into ShrinkIt, start up your system from your hard drive as you normally do. (If you don't usually start up into the Finder, get to the Finder by quitting your startup application or by launching the Finder from your program selector.) Insert the Bonus Stuff disk and double-click its icon. Then click the GSHK icon—that's the GS ShrinkIt application. Now you're ready to install part or all of the Bonus Pack.

We'll run you through installation of the Bonus Desk Accessories; they're typical:

1. Choose “Open” on ShrinkIt's File menu, insert the Bonus Stuff disk, and double-click the file “Desk.Accs.SHK” to open it. The contents of the file will appear in ShrinkIt's archive window (Figure 2C).
2. Pull down the Edit menu and choose “Select All.” The files in the list will turn green.
3. Click the “Extract” button. A Standard Save dialog will appear. Click the “Volumes” button, then double-click the disk you want to install on (your startup disk).

4. When the list of files and folders on your startup disk appears, find the System folder and double-click it to open it.
5. Find the Desk.Accs folder on that list, and double-click it to open it. The name at the top of the window should now read "Desk.Accs" (Figure 2D).
6. Click the "Extract" button and stand back while ShrinkIt places all the Bonus Desk Accessories in the Desk.Accs folder on your startup disk.
7. When that's finished, select "Close" from the File menu.

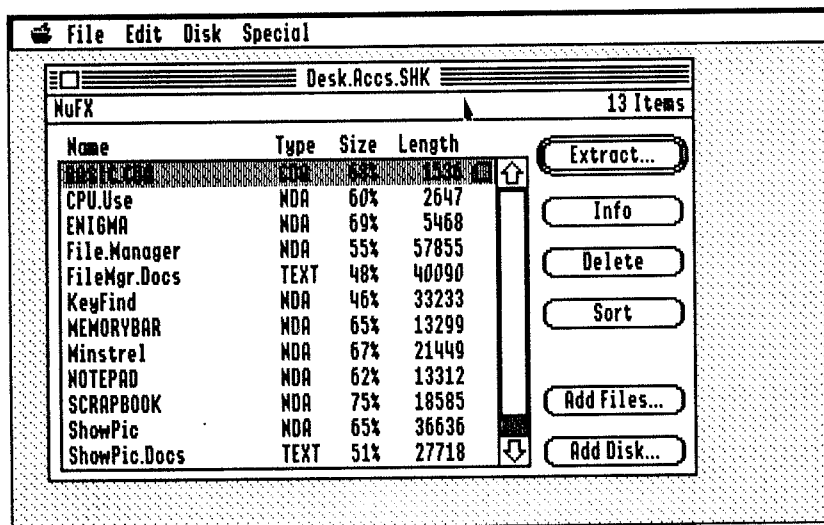


Figure 2C—ShrinkIt: Archive Window (Desk.Accs.SHK Opened)

Once you've done that, you'll have a pretty good handle on how to install most of the other Bonus Pack items. Here are some pointers:

- **Sounds:** Before you install the Bonus Sounds, make sure you've already installed the Sound control panel (see previous section). There are two disks worth of sounds. You'll insert the Bonus Sounds 1 disk first and open the "Sounds.1.SHK" file. Then choose "Select All" from the Edit menu, and click "Extract." Follow Steps 3 and 4 above. In Step 5, open the Sounds folder (not Desk.Accs), then click Extract and stand back. Now close the Sounds.1.SHK archive ("Close" from the File menu). Follow the same procedure with the "Bonus Sounds 2" disk and the "Sounds.2.SHK" file. You won't need to tell ShrinkIt to move to the Sounds folder when you're unpacking the second batch of sounds—it will remember that folder from the first extraction.

- **Fonts:** There are also two disks of Bonus Fonts. They go in the Fonts folder inside the System folder on your startup disk. Follow the instructions for "Sounds" above, except remember that wherever it says "Sounds" you should think "Fonts."
- **Finder Extensions:** The Bonus Finder Extensions are on the Bonus Stuff disk, in a file called "Finder.Ext.SHK." They go in the System.Setup folder inside the System folder on your startup disk. Follow the instructions above for installing the Desk Accessories, except open "Finder.Ext.SHK" in Step 1 and open the System.Setup folder in Step 5.
- **Icons:** The Bonus Icons are also on the Bonus Stuff disk, in a file called "Icons.SHK." Open this file as above. However, in Step 4, open the Icons folder instead of the System folder. Skip step 5 entirely—the Bonus Icons go directly in the Icons folder on your startup drive.
- **Clip Art:** The Bonus Clip Art is on the Bonus Stuff disk in a file called "Clip.Art.SHK." Open this file using Steps 1 through 3 above, using the appropriate disk and file. *Skip Steps 4 & 5*—unpack the clip art to your startup disk by clicking "Extract" in the Standard Save dialog. ShrinkIt will automatically create a folder called Clip.Art and place all your Bonus Clip Art files in this folder.

There, you're done (whew)! Choose "Quit" from ShrinkIt's File menu to return to the Finder, then restart the computer to take advantage of the new Bonus Pack features!

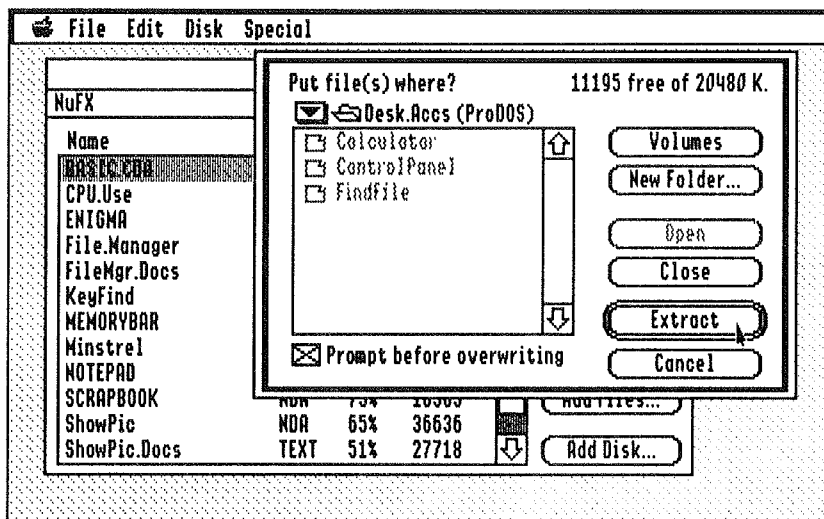


Figure 2D—ShrinkIt: Extracting Bonus Desk Accessories to Desk.Accs Folder

**Note!**

If you don't want to install all of the Bonus Desk Accessories (or Sounds, or whatever), check the disk for a ShrinkIt instruction file. Among other things, this will tell you how to extract only some of the files from a ShrinkIt archive. (Basically, you click only the file you want to install instead of choosing "Select All.") Plus you'll learn how to use one of the most useful IIGS programs around! Alternately, you could unpack the Bonus files to another folder, and move the ones you want to use into the appropriate folder inside your System folder.

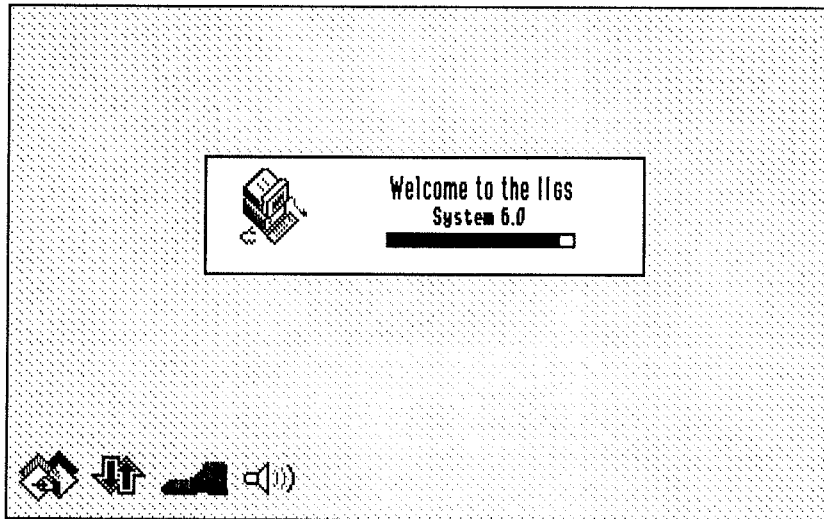


Figure 2E—System 6 “Splash Screen”

Starting Up System 6

You'll start up System 6 the same as any other disk—by turning the computer on with the appropriate disk in a drive (or with your hard drive ready to go), or by holding down ⌘ and Control while pressing and releasing the Reset key. And, on the surface, System 6 looks a lot like System 5, which you may already be familiar with.

The first thing you see is the System 6 “splash screen” (Figure 2E). This screen tells you that you're using System 6 and gives you some indication of how much longer the system will take to start up, via the red thermometer bar, which fills as System 6 loads.

**By The Way...**

The thermometer fills at a rate that reflects how long it took to load System 6 the last time you started up your computer from that disk. It's only an estimate. Your startup time may change when you add or remove items from the System folder.

thermometer

As System 6 loads, it displays small pictures—icons—across the bottom of the screen. These small pictures tell you which startup items are loading—for example, the “boot” icon tells you that the SetStart control panel has just loaded.

▲ **By The Way...**

You can keep your startup items (called “inits” in IIGS parlance) and desk accessories from loading by holding down the Shift key as you start up the computer. When the message “No Inits/DAs” appears under the “Welcome to the IIGS” banner, you can release the Shift key.

When the thermometer fills and the System 6 splash screen disappears, a program called the Finder usually appears. (You can change your startup program with the SetStart control panel). The Finder is “home base” for IIGS users—you’ll use the Finder to back up and maintain your disks, as a “launch pad” for running your programs. If you’re unfamiliar with the Finder (or wish to learn more), be sure to read Chapter 5.

Switching Between System 5 & System 6

Once System 6 has been installed, you can switch back to System 5 without removing System 6 by doing the following things (assuming you took our suggestion about renaming your existing Icons and System folders):

- Open your startup drive’s icon.
- Click the System folder, type “System6” and press Return.
- Click the Icons folder, type “Icons6” and press Return.
- Click the System5 folder, type “System” and press Return.
- Click the Icons5 folder, type “Icons” and press Return.
- Restart your IIGS by holding down the ⌘ and Control keys while pressing and releasing the Reset key.

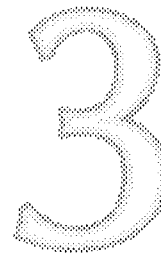
You can switch back to System 6 from System 5 (assuming you’ve followed the above procedure) by performing these steps:

- Open your startup drive’s icon.
- Click the System folder, type “System5” and press Return.
- Click the Icons folder, type “Icons5” and press Return.
- Click the System6 folder, type “System” and press Return.
- Click the Icons6 folder, type “Icons” and press Return.
- Restart your IIGS by holding down the ⌘ and Control keys while pressing and releasing the Reset key.

Don't Forget!

Check the System 6 disks for "Read Me" files that contain last-minute instructions that didn't make it in time for the manual. The "Shortcuts" file on the System Tools 2 disk is full of interesting information, some of which you won't find anywhere in this book!

And... remember to *enjoy System 6!*



The Desktop Metaphor

In the early eighties, some software engineers at Xerox's Palo Alto Research Center (PARC) began working on more effective ways to allow people to communicate with computers. Their studies found that most people process pictures faster than words, and that they prefer to manipulate their work as directly as possible. Based on these results, they began designing a brand new user interface that was intuitive, pictorial, and direct. (The user interface is how you, the user, communicate with your computer. Early user interfaces were based on typing commands. Today's Graphic User Interfaces, also called GUIs, owe a lot to Xerox's research.)

Steve Jobs, co-founder of Apple and father of the Macintosh, saw an early version of the Xerox team's user interface on a visit to PARC. He decided that the basic elements of the Xerox user interface (an on-screen pointer linked to a pointing device called a mouse, along with movable windows, small pictures called icons, and a compact pull-down menu system) would be a part of Apple's next computer—the Lisa.

The Lisa, which cost \$10,000, was a less-than-stunning commercial success, so Apple soon dumped it and introduced a similar but less expensive computer—the Macintosh. The rest of *that* story is, of course, history—and the technology also made its way to the Apple II line. The Apple II GS is the first Apple II to feature built-in support for the Apple Desktop user interface.

So What's The Metaphor?

For keeping cows in, of course. But seriously, a metaphor (when you're talking about computers—forget the definition you learned in English class) is simply a computerized model of the real world. Most people work at a desk; the desktop metaphor is a way of allowing you to do work on the computer in a manner as close as possible to the way you work in the real world, using simulations of familiar tools.

Of course, the simulation isn't perfect. No one who works with a computer *really* thinks they're working on a desktop. Most people's desks aren't the bright blue of the IIGS Desktop, for one thing, and when writing notes on a real notepad most people don't reach for the Edit menu to select Cut when they make a mistake. But the Apple Desktop has been tested and refined to such an extent over the years that the IIGS is probably as easy to use as a computer can be—at least until the next user interface breakthrough. The Desktop's goal is to make the computer an extension of your mind, so you can work without thinking about how to use the computer.

Consistency

One of the real conceptual breakthroughs of the Apple Desktop is that all Desktop programs behave similarly. When you see the blue desktop screen appear with the menu bar at the top, you can assume that the program does certain things in certain well-defined ways, *even if you've never used that particular program before*. For example, if you want to stop using the program, you'll always find the Quit command in the File pull-down menu. If you want to move some of your data from one part of a document to another, you first select the data you want to copy, then you use the Cut and Paste functions in the Edit menu. *Always*. Furthermore, you can safely assume that ⌘-X activates the Cut function and ⌘-V activates the Paste function.

We'll cover all of the above information in more depth later. For now, just remember that consistency is an important feature of the Apple Desktop. Individual programs may extend the desktop metaphor in ways that make sense in that program, but every true Desktop program is, to a point, similar to every other Desktop program.

Object-Action Orientation

When you're working on something in the real world, usually you first decide what you're going to work with, then you decide what you're going to do. For example, if you're proofreading your latest best-seller, you'll probably notice a misspelled word *before* you reach for your red pencil to correct it.

The Apple Desktop works the same way. In the Apple Desktop, you always select the thing you want to work on (the object) before telling the computer what to do with it (the action). For example, if you want to remove a 3.5" disk from the Finder Desktop, you'll first click on the disk, then select "Put Away" from the File menu. If you want to copy a word from a word processing document to the clipboard, you'd first select the word (by double-clicking it), then select "Copy" from the Edit pull-down menu. This object-action orientation is consistent in Desktop programs, and it's designed to make using the IIGS more like the way you really work.

Contrast this with the way programs like AppleWorks work. First you have to decide you want to delete something, then you tell it what to delete. If you change your mind about the action you want to perform, you have to cancel the operation and choose a new one. Then you have to re-select the object you want to work with—even if it's the same one as before.

In the most frequently used operations, the action and the object merge. For example, double-clicking an icon in the Finder opens it. The first click selects the object and the second click is the action, but they're so closely linked that they're one intuitive operation in the user's mind. This is the Apple Desktop at its best!

Split Personality IIGS

The Apple IIGS is, uniquely, designed to run both IIGS Desktop programs *and* older Apple II-compatible (8-bit) programs. It's possible to run both a friendly Desktop program like the Finder *and* an older program with no user interface to speak of *on the same machine!* The authors of this older Apple software had to design and program their own user interfaces because Apple II models before the IIGS didn't have built-in support for the Apple Desktop. The result: no two programs ever worked the same way. (The consistency factor, remember?)

Some newer programs for 8-bit computers (Apple IIe and IIc) *do* use the Apple Desktop user interface. The designers of these programs took it upon themselves to make their programs work like the Apple Desktop, even though it meant programming a complete user interface. Although *Publish It! 4* and *ProTERM 3.0* (to name two examples) run on the IIGS as well as older Apples, and although they *work* somewhat like Desktop programs, *they're not IIGS Desktop programs.*

When you're using any 8-bit program on your IIGS, your IIGS is essentially pretending to be a IIe, and most of the special features of the IIGS are not available. You won't have access to System 6 features like New Desk Accessories (NDAs), File System Translators (FSTs), and IIGS printer drivers because, in its IIe emulation mode, the IIGS cannot support these features. That's not to say these look-alike programs aren't useful—many are. Just keep in mind that most of System 6's features work only from *real* Desktop programs.

▲ By The Way...

If there's a rainbow-colored Apple at the left end of the menu bar, you're probably using a real Desktop program. If there's no Apple or if the Apple isn't rainbow-colored, you're probably using a program that only looks like a Desktop program.

A Closer Look

Let's take a look at a screen from a IIGS Desktop program (Figure 3A). This screen happens to be from the Finder, but other Desktop programs have a similar look and feel. (The Finder is a bit unusual because it is specifically designed not for a particular application like word processing, but rather as an organizational tool for your disks, documents, and programs. Still, it's a program you'll be using frequently, so it's a good place to start.)

You'll first notice the menu bar at the top of the screen. The menu bar is the Desktop's "main menu." In non-Desktop programs, you're usually presented with a main menu containing general things you can do, which branches out into submenus. For example, in AppleWorks, when you select "Add Files To Desktop" from the main menu, you're presented with a submenu that allows you to create a new file or load an existing file to the Desktop.

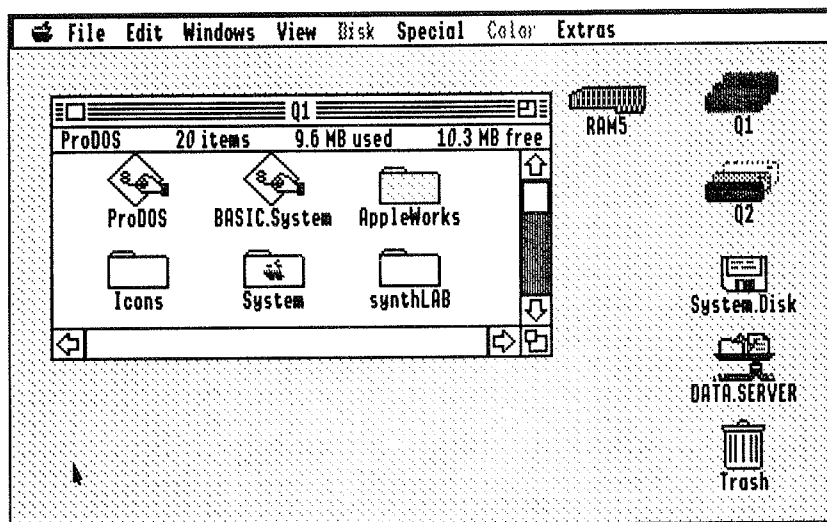


Figure 3A—Sample Desktop Screen (from Finder)

▲ By The Way...

The AppleWorks desktop, in case you haven't guessed, is not at all related to the desktop metaphor we're talking about in this book. The AppleWorks desktop is just a place to store your files while you're working on them. The Apple Desktop is a much more general concept—it's a way of interacting with your computer.

The Desktop's menu bar is its main menu—it lists all the *kinds* of things you can do in the program you're using. In the Finder, the things you can do are divided into the categories of File, Edit, Windows, View, Disk, Special, and Color. The rainbow-colored 🍏 symbol to the left of the File is also a menu. We'll explain how to use the menus and the options in them a little later; for now, it's enough to realize that the menus are there.

The menu bar is almost always visible in a Desktop program. That's one of the basic ideas of the Desktop—to make it clear at all times what you can do next. The menu bar takes up very little space on the screen when you're not using it, but you can glance at it (or even browse through it) at any time to remind yourself what options are available.

Another fundamental Desktop element is the window. A sample Finder window occupies most of the middle of the screen. The Finder uses windows to represent the contents of a disk. Like most Desktop windows, the Finder's windows can be moved, resized, opened, closed, and scrolled. You can also open more than one window. (We'll get into window manipulation shortly.) What you see in a window is called, naturally, a view.

Icons are yet another Desktop element. Icons are small pictures representing documents, programs, and other things. On the sample desktop, you can see icons for two hard drive partitions (named Q1 and Q2), a RAM Disk (named RAM5), a 3.5" disk (System.Disk), and an AppleShare network volume (DATA.SERVER). There's also a Trash can, which is used for removing things from the Finder desktop. Additionally, the Q1 window contains even more icons, representing documents and programs on the hard drive.

Finally, find the small arrow in the lower left hand corner of the screen. This is the pointer (or cursor). When you move the mouse, this small arrow moves correspondingly on the screen. You use the pointer to select icons, manipulate windows, and choose items from pull-down menus. In fact, our very next chapter is all about using the mouse—so stay tuned!

▲ **By The Way...**

The elements of the Desktop (windows, icons, menus, and pointer) happen to form an acronym—WIMP. The term was originally coined by expert computer users who actually resented the simplicity of graphic user interfaces like the Apple Desktop. These folks were used to typing commands—and they liked it! These days, the term WIMP is used in a merely descriptive fashion to refer to programs or computers that use a graphic user interface.

Mouse Basics

This chapter is for IIGS users who have limited experience with the IIGS mouse. Although such a chapter may seem unnecessary—most everyone has used a mouse at one time or another—this chapter also ensures we’re all talking about the same thing when we talk about mouse actions.

The mouse is a simple, elegant device for moving the pointer around on the IIGS screen. On the bottom of the mouse is a small rubber ball, which rolls as you move the mouse around your desktop. Sensors inside the mouse track the rotation of the ball, and, like magic, the on-screen pointer follows your hand movement.

The mouse only detects movement when it is on a smooth, flat surface. If you run out of desk space before you reach the spot you’re aiming for on the screen, you can just pick up the mouse and move it back. Since the ball isn’t touching the desk, the mouse doesn’t register the movement, and you now have more room to move in the direction you want to go. (Experienced mousers pick up the front end of the mouse just enough to move it without the ball touching the desk.)

The mouse also has a long tail-like cord connecting it to the computer, and a button on the top. (If the cord is the mouse’s tail, the button is positioned atop, well, its hind end.) This button is used to select documents, icons, and menus. You may have seen mice on other computers with two or three buttons—but you do can everything with a single button on the Apple Desktop.

If you’re just getting started with the mouse, you may want to start up the System Disk to get to the Finder and follow along.

Pointing

The simplest mouse action is pointing. To point to something on the screen, just move the pointer to it. Position the pointer on top of the object you wish to point to—not above it, but literally on top of it (Figure 4A), as if you were looking down on the desktop from above and the pointer was on top of the object. (In Figure 4A, we’re pointing to the System Disk. Look closely at the System Disk icon to find the pointer.)

When pointing to extremely small objects, like the small box in the upper left hand corner of a window, it's useful to know that the arrow's "hot spot" is at its tip. If the tip of the arrow is inside the box, you're pointing to it. If the tip is out of the box, you're not pointing to it—even if some other part of the pointer is in the box. So when we say to point to an object, we mean to put the tip of the arrow on it.

By itself, pointing isn't too useful—it merely tells the computer what object you want to work with. To actually start working with an object, you need another mouse action—clicking.

Clicking

Clicking is simple—a quick press and release of the mouse button. Clicking does different things depending on what program you're in and where you click. In the Finder, clicking an icon turns it black (we call that "highlighting" or "selecting" the icon). Clicking various parts of windows also has interesting results. Clicking on text in a document moves the insertion point to the clicked location (don't worry, we won't talk about the insertion point until Chapter 5, so you haven't missed anything). Clicking on a button in a dialog box activates that button. And so on.

By the way, in Desktop-speak, you don't click *on* an object, you just click it. (You don't "click the mouse on the Q1 icon," you just "click the Q1 icon.") This usage encourages you to think of the mouse and the on-screen pointer as being not just linked but literally *the same thing*.

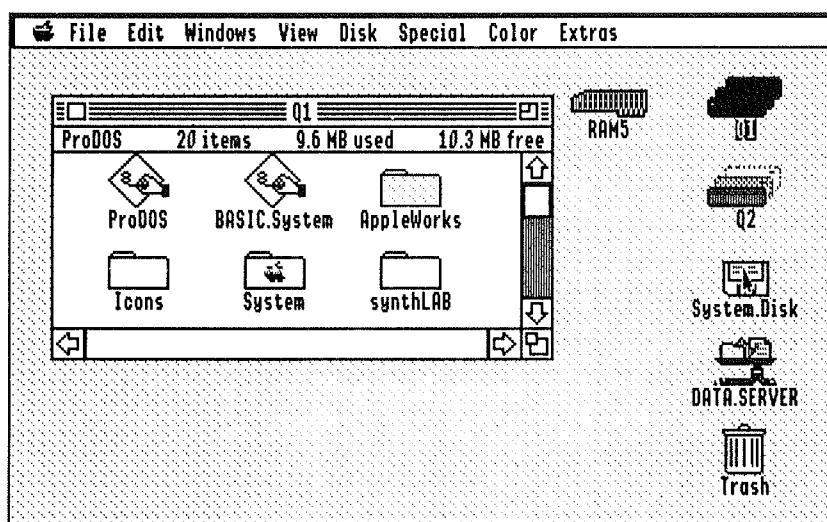


Figure 4A—Pointing to an Icon in the Finder

Pressing, Dragging, & Dropping

Pressing

Pressing the mouse button and holding it down is called, logically enough, pressing. As with clicking, you don't speak of pressing the mouse button; you speak instead of pressing the object you're pointing to.

Pressing a menu title in the menu bar pulls down (displays) the menu. If you're following along in the Finder, try it with the File menu. Pressing, alone, is useful to see what's on a menu without actually selecting an item. Release the button to remove the menu from the screen.

Dragging

To drag an object, continue to hold down the mouse button while moving the mouse. You'll use the drag technique with icons (try dragging the System Disk icon around) and pull-down menus. For now it may not seem too useful to drag icons around on the Desktop, but that's how you copy files from one disk to another (more on that later).

Sometimes you'll be required to "drag a rectangle." You use this technique in the Finder to select a group of icons that are close together on the screen, and to draw boxes in paint programs. To drag a rectangle, you press the mouse at one corner of the rectangle you want to define (it doesn't matter which corner you start with, though most people start with the upper left corner out of habit) and drag it to the corner *diagonally opposite* the corner you started from (the lower right corner in our example). Try dragging a rectangle around some of the icons on the Finder screen if you're following along.

Dropping

Dropping is the action that occurs when you release the mouse button after a drag. (Sometimes this is simply called releasing, too—you generally "release" menus.) When you're dragging an icon, you'll probably want to drop it eventually. It's natural to speak of dropping it back onto the desktop, or into a folder, or into the trash.

Other Mouse Maneuvers

Double-Clicking

Double-clicking is two quick clicks of the mouse with very little delay between the two clicks. The computer registers these two clicks as one event. In the Finder, double-clicking an icon opens it (a useful shortcut for selecting Open from the File menu); in most word processors, double-clicking a word selects that word (remember, single-clicking in a document just moves the insertion point). Occasionally you'll even be called upon to perform *triple*-clicking.

▲ **By The Way...**

You can adjust just how the computer responds to double-clicks in the General Control Panel. If you tend to double-click very quickly, but the computer is expecting you to double-click slowly, you may find it occasionally registering double-clicks when you don't mean them. If you double-click slowly, and the computer is set for fast double-clicks, the computer may interpret your double-clicks as two separate clicks.

Shift-Clicking

Shift-clicking means you hold down the Shift key on the keyboard while clicking. This is often used to select a group of icons, or a number of items from a list, or a series of words in a word processor. There's also Apple-clicking, Command-clicking (the same as an Apple-click—that weird propeller symbol is Apple-speak for "command"), Option-clicking, and Control-clicking; different programs define these in different ways.

Cursors

The IIGS uses several different cursors to give you clues about what various mouse actions will do in a particular program. A cursor is just a more generic word for the on-screen pointer you're already familiar with—the arrow is just one cursor, though it's probably the most common one. The cursor changes when you move it from one type of area into another on the screen—for example, from an arrow to an I-beam when you move it into a text area and back to an arrow when you move it out of the text area. Visual feedback like this always lets you know what will happen when you click (or drag, or double-click, or...) and lets you know instantly when a program expects a certain kind of action.

The Arrow



The Arrow cursor appears when you're selecting items from a menu, when you can click on buttons in dialogs, and in places like the Finder where you can select and double-click icons. (In some drawing programs, you can select an object in your drawing with an arrow cursor.) Single-clicking selects an object; shift-clicking is used to select more than one object; double-clicking opens the object. In many programs (like the Finder) you can drag a rectangle around a group of objects to select them all. The arrow's hot spot—the point that determines where the computer thinks you're pointing—is at its tip.

The I-Beam



The I-Beam cursor appears in word processing documents and whenever the cursor is over a part of the screen where text can be entered. Clicking the mouse positions the insertion point; shift-clicking selects text from the insertion point to the cursor; double-clicking selects a word. The i-beam's hot spot is in the center of the vertical bar.

The Plus



The big plus-sign cursor (*not* the puny crosshairs you frequently see in graphics programs) appears in spreadsheets and other programs that manipulate grids with data in them. Clicking one data element (a cell) selects that cell; shift-clicking selects from the current cell to the cursor; double-clicking often opens a cell for editing. The hot spot is in the center of the cursor.

The Watch & The Beach Ball



The IIGS displays the wristwatch cursor when it's doing something that will make you wait for a moment. It's the visual equivalent of a "Please Wait" message. The rotating beach ball cursor means pretty much the same thing—except for the minor difference that your computer is out having fun at the beach while it's displayed.



The Pointing Finger



The HyperCard and HyperStudio programs use this cursor to allow the user to click on buttons on cards. Although HyperCard and HyperStudio allow you to have icon-like pictures in a stack, you don't double-click them to activate them (a single click is sufficient)—therefore, an arrow cursor is inappropriate. A few other programs may also use this cursor to indicate the single-clicking, not double-clicking, is used for activating icons. The hot spot is at the finger's tip.

Custom Cursors

Some programs have cursors of their own other than the ones we've described here. Paint programs, in particular, have a wealth of unique cursors which remind you which tool you're using. If you see an unfamiliar cursor, check the program's documentation to see what it's for.

Pull-Down Menus

Using The Menu Bar

All Desktop programs feature a menu bar. The pull-down menus, accessed by pressing the menu titles (Apple, File, Edit, etc.) are used for selecting the actions you want to perform and for selecting options that affect the operation of the program. In the old days, menus took up the whole screen (or a good portion of it) and you chose options from menus by typing numbers and letters or by using the arrow keys to move a cursor from one item to the next. The Desktop's menu bar doesn't require any typing, and it puts all your options literally at your fingertips.

All Desktop menus work the same way, so once you've learned how to operate the pull-down menus in one program, you've mastered them all. We'll be using the Finder for our examples again, just because it's easy to get to. Boot your System Disk and follow along if you like.

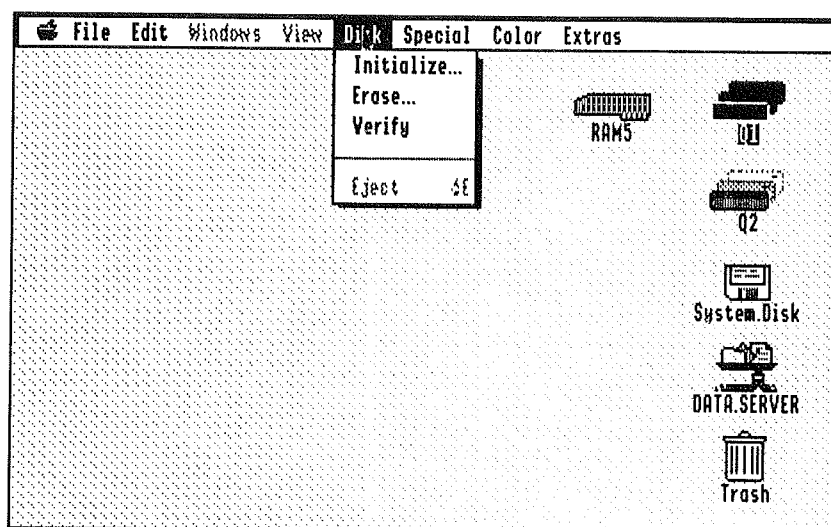


Figure 5A—Pull-down menu (from Finder)

To look at a pull-down menu in a Desktop program, just press the title of the menu you want to look at (Figure 5A). The menu will appear and its title (e.g. "Disk") will be highlighted (white letters on a black background). To dismiss the pull-down menu, release the mouse button. If you just want to browse through the available pull-down menus, continue pressing as you move the mouse left and right across the menu bar. Menus will pull down and pop up as you move the pointer across the menu titles.

Selecting an item from the pull-down menus is just as simple. Press the menu title and continue to hold down the mouse button as you drag downward through the items on the menu. As the pointer moves down, the items you point at will be highlighted. Once you've found and selected the menu item you want to use, just release the mouse button. The menu item will flash to signify that your choice has been accepted, and the menu will disappear.

If you decide you don't want to select anything from the menu after all, just move the pointer off the menu entirely and release the mouse button. Or, to select an item from another menu, continue pressing the mouse button and point to one of the other menu titles on the menu bar.

Dimmed Menus & Items

Some menu titles appear dimmed. This indicates that none of the options on the menu are valid actions in the program under the current circumstances. For example, in Figure 5A, the Windows menu is dimmed because no windows are open. Individual items within a menu can also be dimmed—for example, the "Eject" option in the Finder's Disk menu will be dimmed if the disk icon you've selected can't be ejected (if it's a hard drive, for example).

Checked Menu Items

Some menu items have a check mark beside them. For example, the Finder's View menu (Figure 5B) contains a checked item which indicates how the icons in the window are organized. In our example, "by Icon" is checked. If we selected "by Name" from the menu, the window would change to display a list of the icons organized alphabetically by name and the check mark would move to the "by Name" selection. In this case, we can use the menu for two things—to actually change the view of a window, and to see which view a window currently has by looking for the check mark.

The check mark is used in many other programs. For example, word processors have a check mark in their Font menu to tell you which font (type style) you're using, as well as check marks beside each style (bold, underlined, etc.) and size (10 point, 12 point, etc.) in use.

In some cases, the check mark is used in a slightly different way—for example, in a game, there might be a menu item called "Sound." If this item is checked, the game makes noise; if it's not checked, the game is silent. Selecting the "Sound" menu item toggles the check mark—turns it on if it's off, or off if it's on. The context in which the check mark is used will help you determine whether it's used in this way, or in the first way of choosing one choice from a number of possible choices.

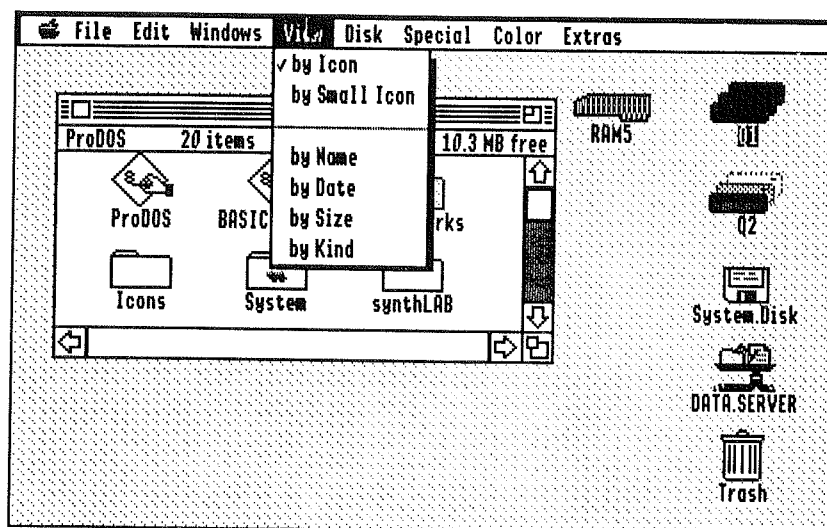


Figure 5B—Check marks and menu items (from Finder)

Using Long Menus

Some menus have far too many options to fit on one screen. Platinum Paint (Figure 5C) is a good example of a program with so many features that the author was hard pressed to cram them all on one screen.

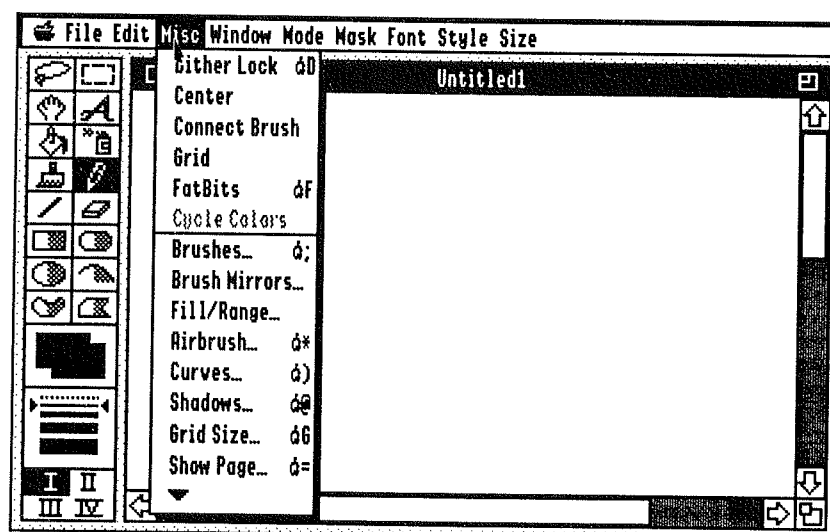


Figure 5C—Long Pull-Down Menu (from Platinum Paint)

When you pull down, for example, the Misc menu, a small downward-pointing arrowhead appears at the bottom to let you know there are more items than can be displayed. Drag the pointer to this arrowhead and the menu will begin scrolling—moving the displayed menu items upward and revealing new items at the bottom. An up-arrowhead appears at the top of the menu; use it to scroll in the other direction and re-display the items that scrolled off the top.

Once you've found the item you want, selecting it works the same as if the menu didn't scroll. Just point to the item and release the mouse button.

Ellipsis...

Menu items followed by an ellipsis (three periods) always lead to a dialog. Dialogs are used to specify information the computer needs to complete your request. For example, if you choose to initialize a disk in the Finder, you'll need to tell the Finder the name for your new disk, as well as the format you want to use. A dialog appears to request this information when you select "Initialize" from the Disk menu. (We'll be talking about dialogs shortly.)

The Standard Menus

The first three menus on the menu bar are the same in nearly all Desktop programs. These are the  (Apple) menu, the File menu, and the Edit menu. (The rest of the menus will be tailored to each program's purpose.)

The  menu usually contains an "About" item which displays information about the program you're running (its name, its author, the version number, and so forth). In many cases the  menu also includes an option for on-screen help or other assistance. Below these menu items (if any), the  menu contains your New Desk Accessories (NDAs). NDAs are available from any Desktop programs and are like real-life desk accessories—calculators, note pads, etc. We'll be talking more about NDAs later.

The File menu contains options for creating, opening, saving, and closing documents (to work with a document, you open it first; when you're done, you close it). You'll also usually find a few other file-related options in the File menu, like "Print," as well as the program's Quit option.

The Edit menu contains items which allow you to change the information you've selected in some way. The "Cut," "Copy," and "Paste" options are used to transfer information from one document to another using a lingo feature called the clipboard. The "Cut" option deletes the selected information from the document and places it on the clipboard. The "Copy" option leaves the information in your document but places a copy on the clipboard. The "Paste" option places the contents of the clipboard into the document you're working on. The clipboard can be used not only to exchange information between documents but also between programs.

The Edit menu usually also includes an "Undo" option, which reverses the effect of the last action you performed. (Often selecting "Undo" again after undoing an action will "re-do" that action). There's also usually a "Select All" option, for selecting everything in the document you're working on, and a "Clear" option, for deleting the current selection.

Key Equivalents

In most programs, the most frequently used menu items are also available through keys on the keyboard. Look for menu items with the ⌘ symbol next to them, like Eject in Figure 5A. Holding down the ⌘ key while pressing the letter E is the same as selecting “Eject” from the Disk menu. As you become more familiar with the operation of the programs you use the most, you’ll find yourself reaching for the key equivalents more and more—they really are time-savers, especially in programs like word processors where it can be a hassle to continually move your hand from the keyboard to the mouse and back.

➡ Shortcut

The following key equivalents are supported by many (but not all) Desktop programs. The first four Edit menu key equivalents are the first four keys at the left of the bottom row of letters—Apple chose them because they weren’t as likely to be needed by programs for other functions.

File menu		Edit menu	
Open	⌘-O	Undo	⌘-Z
Close	⌘-W, ⌘-K	Cut	⌘-X
New Document	⌘-N	Copy	⌘-C
Save	⌘-S	Paste	⌘-V
Print	⌘-P	Select All	⌘-A
Quit	⌘-Q	Clear	Delete

Additionally, virtually all Desktop programs support ⌘-Period and/or Escape to cancel or stop an operation in progress.

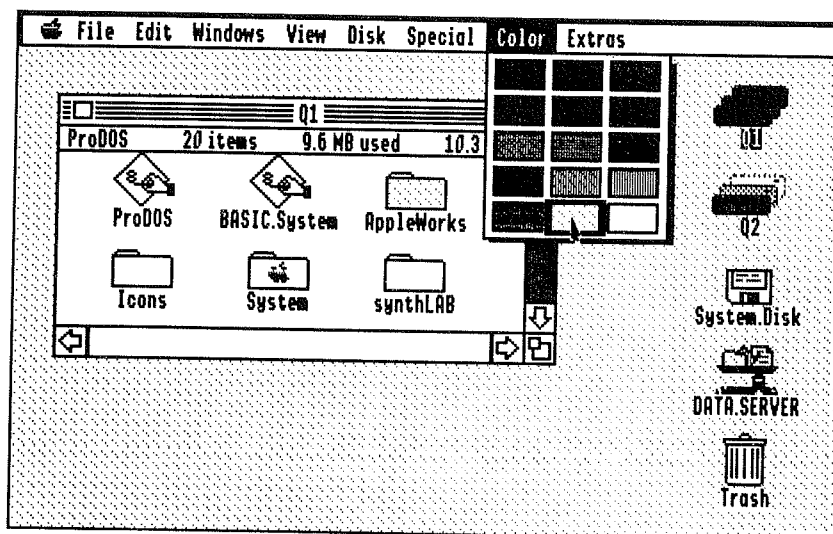


Figure 5D—Custom Menus (Color Menu, from Finder)

Custom Menus

Some programs include menus that are somewhat different from the normal pull-down menus. A good example is the Color menu in the Finder (Figure 5D). Pulling it down reveals not a list of colors, but the colors themselves in a five-by-three grid. You select a color from this menu the same way you'd select an item from a regular menu, except you can move the pointer sideways. Some other programs—like HyperStudio—display small icons for their tools in similar menus.

Tear-off menus are gaining popularity. In some programs, like HyperCard GS, you can detach a menu from a menu bar and leave it lying on the screen. This is especially useful with pull-down color menus and tool menus. Once menus are torn off, they become windoids which look and behave somewhat like regular windows except that they stay in front of any other windows on the screen. (Don't worry, we'll explain windows and windoids in due course.) You can then select an item from the torn-off menu simply by clicking on it, and you can move it around the screen wherever you want. You can dismiss the menu by clicking the windoid's close box.

To tear off such a menu, pull it down as usual. Then drag the pointer outside the menu. An outline of the menu will follow the pointer. Release the mouse button and the menu appears on your screen where you dragged it. Remember, this trick doesn't work in all programs, just the ones which are specifically designed to support it.

Note!

The Finder's Color menu does not tear off. Attempting to do so will only make you feel silly.

Hierarchical menus are used frequently on the Macintosh but are rarely seen on the IIGS. In fact, we only know of one program that uses them, and neither the IIGS nor System 6 support them directly, so you probably won't see very many of them. But nevertheless, if you see a rightward-pointing arrowhead to the right of a menu item, you'll know you're about to encounter a hierarchical menu. When you highlight that menu item, a second menu will "pop up" to the right of the arrowhead. Slide the pointer into this submenu and select an item from there (or move the pointer back into the pull-down menu to get rid of the pop-up menu).

Alerts & Dialogs

Using Alerts

Alert windows (or just "alerts") appear when something unexpected has happened to find out what you intend to do about it. Take, for example, Figure 5E, where we have foolishly told the computer to move the program "Teach" into a folder which already contains a program by that name. (Well, actually, we might have meant to do that. Alerts don't necessarily mean you've made a mistake.) In any case, the computer doesn't know whether or not we meant to replace the existing "Teach" program, so it asks us to confirm our action before erasing the old file.

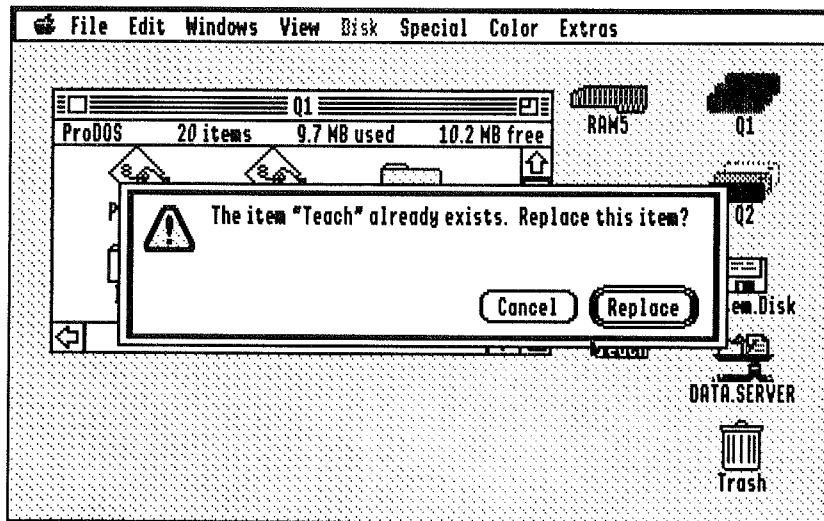


Figure 5E—Sample Alert (from Finder)

The alert window contains three important elements. First, there's an icon, which indicates what level of alert you're looking at (Figure 5F). The three alerts you'll encounter most often are:

- *Note alert.* The little talking guy with the flat face serves to tell us that the computer is just telling us something that it thinks we should be aware of. We don't necessarily need to take any action, just acknowledge that the computer's message is heard and understood. For example, when you shut down the computer using the Finder's "Shut Down" option (on the Special menu), you get a Note alert that tells you it's OK to turn the computer off.
- *Caution alert.* This is the type of alert we've encountered in Figure 5E. It indicates that there is a possibility of accidentally erasing something you didn't mean to erase (or otherwise messing something up), so the computer has decided to ask you to make sure you really want to do what you said you wanted to do.
- *Stop alert.* With its distinctive red icon and white hand, the Stop alert is the computer's way of saying "You can't do that." For example, if you try to remove a file that's in use by the System software, a Stop alert says "This operation could not be completed because a file on the disk is open." Stop alerts should be considered error messages.



Figure 5F—Alert icons

Then there's the alert message itself. In our case in Figure 5E, that's "The item 'Teach' already exists. Replace this item?" Notice that the message tells us what has happened and also asks us what we want to do.

Using Buttons

We respond via the oval buttons at the bottom of the alert. Click one of these buttons with the mouse to tell the computer what to do. In this case, we can Cancel the operation, or we can Replace the existing program with the one we're moving. A few words have specific meanings when you see them in alerts like this. For example:

- "OK" (or "Okay") means that you acknowledge the information the computer is presenting and that it's all right to continue.
- "Continue" means basically the same thing as "OK" but is usually used in a sense of giving the computer permission to perform an operation after it has asked if it's all right to do so (as opposed to "OK," which is usually just an acknowledgement to a Note or Stop alert).
- "Cancel" means that you want the computer to put things back exactly the way they were before you started the operation, as if you never selected it. For example, if you choose to initialize a disk and then click the Cancel button before the initialization begins, the disk is not initialized and no information is lost. (In the case of the situation depicted in Figure 5E, the move would never happen.)
- "Stop" means to stop the operation in progress. Unlike "Cancel," the operation may be partially complete. For example, if you're doing a search and replace operation in a word processor, you may be able to stop the replace operation, but not cancel it, if you've already accepted some changes. If you see a "Stop" button, but not a "Cancel" button, the computer has already completed at least part of the operation.

Shortcut

One of the buttons usually has a double outline, like the "Replace" button in our example. Pressing the Return key is the same as clicking the double-outlined, or default, button. Pressing - (holding Apple and pressing the period key) or pressing Escape is the same as clicking the Cancel button, if there is one.

⑥ New & Improved

In System 6, you can now click most alert buttons from the keyboard by pressing the first letter of the button's name (excluding double-outlined buttons and the Cancel button, which have their own keyboard equivalents).

Using Dialogs

Dialog windows (or dialog boxes, or just dialogs) are like alerts on steroids, or—if you'd rather think of things the other way around—alerts are stripped-down dialogs. The purpose of a dialog is allow the computer to carry on a kind of “conversation” with the user—that’s you. It’s a way for the computer to ask for information it needs to carry out one of your requests.

For an example, take a look at the Initialize dialog in Figure 5G. It shares some features with the alert we discussed in the last section. There’s a Caution icon (the black exclamation point in the yellow triangle) and the warning that the operation will cause all the information on the disk to be destroyed, and there are the buttons at the bottom of the window. (There’s also a line at the top of the window that tells you the name of the device that holds the disk you’re initializing.)

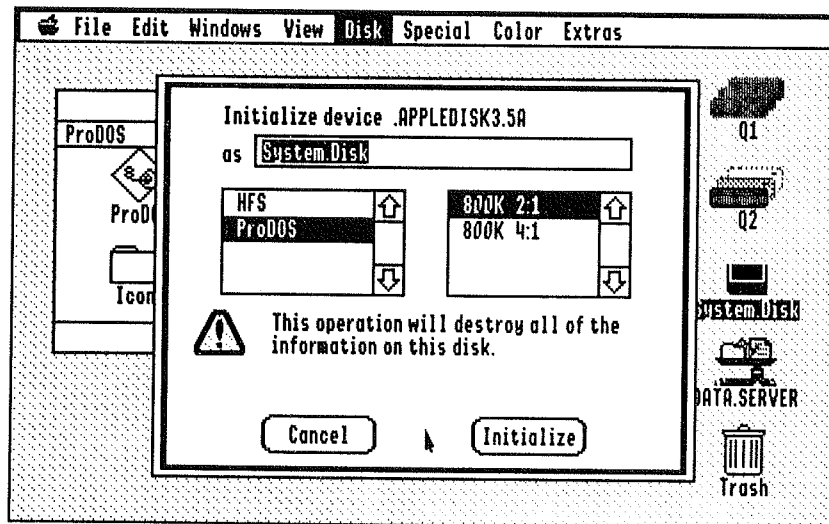


Figure 5G—Sample Dialog (Initialize, from Finder)

There are also three other boxes in the dialog. The topmost one is a place to enter your name for the disk you’re initializing. (This box is filled in with the disk’s current name or, if the disk is blank, with the default name of Untitled.) The box where you enter the disk’s name is called a line-edit field because it’s a place for entering and editing a single line of text; we’ll describe it in detail in a moment.

The two other boxes allow you to select what format the disk will be initialized with (either ProDOS or HFS, the Macintosh disk format) and the disk’s size and interleave. (We’ll leave a discussion of these terms for later.) These two boxes are called lists because they contain a list of your choices. To select one of the items in a list, simply click it. If there are more items in a list than will fit on the screen, you can use the scroll arrows next to the list to scroll the list up and down to reveal more of the items.

Buttons, line-edit fields, lists, and other clickable things in dialogs are known as controls. You set the controls the way you want them, then you activate the dialog by clicking the “OK” button or dismiss it by clicking “Cancel.” We’ll be providing specific examples of how to fill in and use the various dialogs you’ll encounter; for now, it’s enough just to understand in general terms how the parts of a dialog work.

➡ **Shortcut**

System 6 allows you to select items from the lists (and select line-edit fields to type data into) using the keyboard. Press the Tab key to move from one list (or field) to the next; a bold black box surrounds the field you’re editing. Use the up and down arrows to select the options in the list. Hold down the ⌘ key while you press Tab to move to the previous list (or field).

Using Check Boxes & Radio Buttons

The Finder’s Preferences dialog (Figure 5H) shows another type of dialog element—the check box. You generally use check boxes to turn a program’s features on and off—the Preferences dialog is pretty typical. Clicking a check box turns it off if it’s on—marked with an X—or on if it’s off. (It’s the old toggle effect again, which we last saw in the discussion of check marks in the pull-down menus section.)

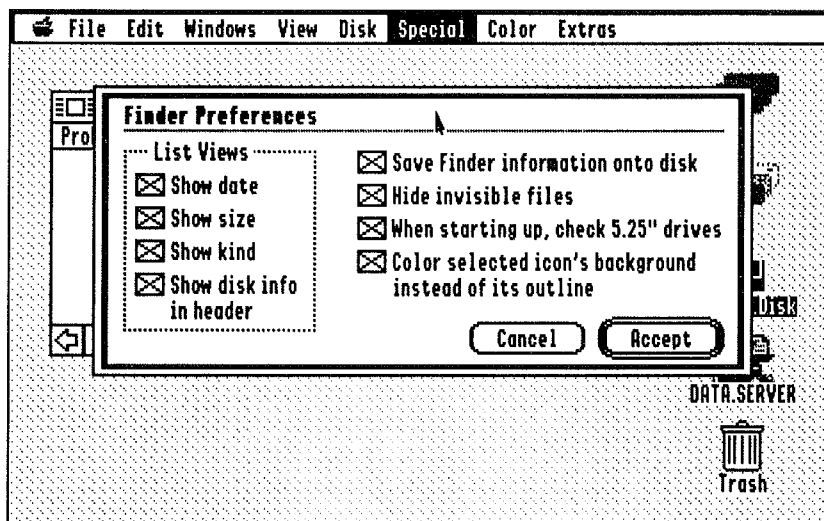


Figure 5H—Sample Check Boxes (Preferences, from Finder)

The Shut Down dialog (Figure 5J) shows the radio buttons—so named because they act like the buttons used to choose your favorite stations on car radios. As with the real radio buttons, you can choose only one option at a time—the options are mutually exclusive: that is, choosing one automatically means that you are rejecting the others. (This makes logical sense—with your car radio, you can’t listen to more than one station at a time, either.)

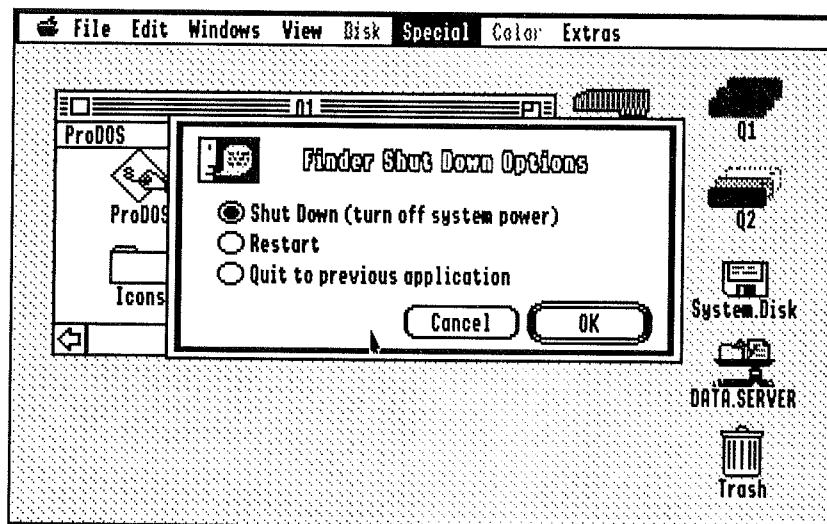


Figure 5J—Sample Radio Buttons (Shut Down, from Finder)

➡ **Shortcut**

In most cases, it's not necessary to click exactly on the check box or radio button; clicking on the text to the right of it is usually just as good (and a lot easier to hit!).

Using Pop-Up Menus

Pop-up menus are often used in place of lists or huge numbers of radio buttons when there's not enough room on the screen to display all the lists or radio buttons. The Finder Help window (Figure 5K) is typical. Though the little box at the top of the window looks like it might be a button, it's actually a menu. It works much like pull-down menus—press it to display the menu, then select an option from the menu by dragging the mouse up or down. The menu's current setting always appears at the cursor position when you first pop the menu up, so if you don't wish to change the menu's setting, simply release the mouse button without moving the mouse.

📌 **Note!**

Pop-up menus often contain more choices than will fit on the screen—especially since the item that pops up under the pointer is whatever option is selected; that can skew the menu toward one end of the screen or the other. The trick for scrolling long menus up and down works just as well with pop-up menus as it does with pull-downs.

▲ **By The Way...**

A downward-pointing arrow (like at the right of the Topic pop-up menu) is often a signal that you're looking at a pop-up menu.

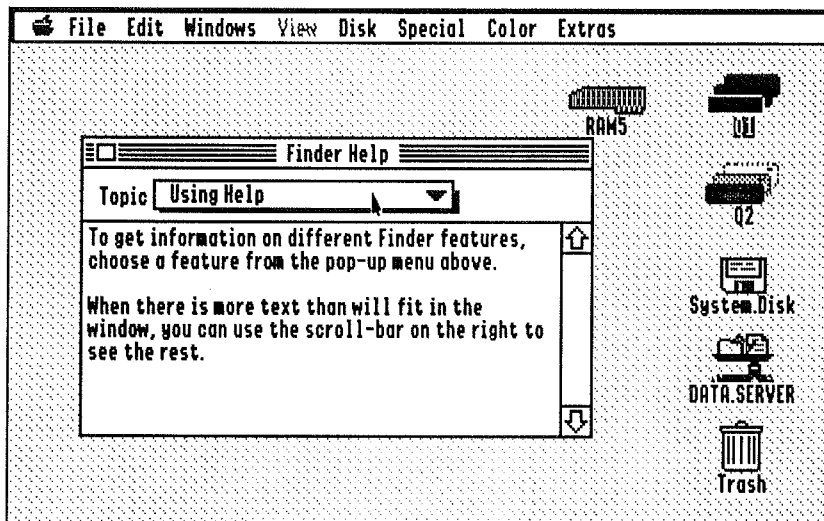


Figure 5K—Sample Pop-Up Menu (Help, from Finder)

Text Fields

Line-Edit Fields

You’ve already been introduced to line-edit fields if you read the section on dialogs above. Line-edit fields are used to fill in text data (such as the name of the disk in the Initialize dialog) that your computer needs to complete a task. Since they’re text-edit fields, you enter data into them by typing. We’ve already discussed how to move from one line-edit field to another using the Tab key (see the “Shortcut” in “Using Dialogs” above).

Text-Edit Fields

Text-edit fields are the multi-line equivalent of line-edit fields. Instead of being limited to a single line of text, you can enter many lines—practically as many as you want, though there’s usually some limit. As with list fields, you’ll often see scroll arrows to allow you to move through a long text-edit entry.

The Finder’s “Comment” card (in the Icon Info window) is a typical text-edit field (although it doesn’t have scroll arrows). Most word processors are based on the conventions of text-edit fields—the window containing your document is considered one big text-edit field.

Both line-edit and text-edit fields react to the same kinds of mouse and keyboard actions, so we’re covering them together. Of course, you’re limited to a single line of text in line-edit fields.

The Insertion Point

An important concept when dealing with any kind of text is the insertion point, which we mentioned in passing in the last chapter. As we noted in that chapter, the mouse cursor changes to an I-beam when it's positioned over a text entry area. Clicking the mouse when it's in a text area moves the insertion point to the cursor position.

So what's the insertion point, anyway? It's a second cursor that's independent of the location of the mouse cursor. The insertion point moves when you type and when you click the mouse on a section of text, as opposed to moving whenever you move the mouse. The insertion point keeps track of where the next key you press will be inserted into your text.

The arrow keys can also be used to move the insertion point around. Additionally, holding down ⌘ and pressing the left or right arrow key moves the insertion point to the beginning or end of the current line; using Option with the left and right arrow keys moves the insertion point to the beginning of the next or previous word.

Selecting Text

To select one or more characters for editing, drag the I-beam cursor across the characters you want to select. (Remember, in the Apple Desktop you first select the object you want to work with—in this case, some text—then you tell the computer what you want to do. Object-action, remember?)

For example, if you want to select the word "her" inside the word "there," you'd first position the I-beam to the left of the letter "h," then press the mouse button and drag across the letters "her" until all three of them are displayed as white letters on a black background. When this happens, release the mouse button. The letters "her" are now selected for editing. (You can also drag backward; the result is the same.)

You can also shift-click to select a section of text from the insertion point to the mouse cursor position. For example, if the insertion point were already to the left of the letter "h," you could simply shift-click after "r" to select "her."

Note!

When selecting multiple lines of text, the selection moves rightward first, then downward. For example, if you click in the middle of the first line of text and drag down to the middle of the third line of text, you have selected the right half of the first line, the entire second line, and the left half of the third line. By the way, most word processing programs will automatically scroll if your selection runs into the edge of the window.

Shift-clicking is also used to extend or contract a selection. If you've already selected text and realize you want to select more (or less) text, hold down the shift key and click the I-beam where you want the selection to end. The selection will grow or shrink as appropriate to where you clicked.

 Note!

What's actually happening when you extend (or contract) a selection is that the computer is keeping track of the original insertion point and adjusting only the end of the selection. It's as if one end of the selection is anchored to the insertion point and you can move the other end around the document as if the selection were a rubber band. Remember, the IIGs recognizes backward drags, so it's possible to move the end of the selection before the insertion point and switch a forward selection to a backward selection!

 Note!

It's also possible to select more text than will fit on the screen (in programs like word processors) by shift-clicking. Click the insertion point, then scroll the window to find the end of the desired selection and shift-click there. (See next section for more information on scrolling.)

Double-click to select an entire word. Triple-click to select an entire line (or the entire field if it's a line-edit field). If you double-click and hold the second click, dragging selects entire words; if you triple-click and hold the third click, dragging selects entire lines.

Selection can also be performed from the keyboard. Use the left and right arrow keys to move the insertion point to the beginning of your selection. Now, hold down the Shift key and press the right arrow key until the desired selection is highlighted. (You can also start at the end of the selection and use Shift with the left-arrow key to "drag backward" with the keyboard.) And, of course, you can also use the ⌘ and Option keys with Shift to extend the selection to the beginning or end of a line or to the previous or next word.

Finally, when you use Tab (or ⌘-Tab) to move from one field to another in a dialog, the field you tab into is automatically selected. (When a dialog first appears, the first field is selected.)

 Shortcut

A quick way to select the entire contents of a line-edit field is to press ⌘ and Left Arrow (to move the insertion point to the beginning of the line), then press ⌘, Shift, and Right Arrow (to extend the selection to the end of the line).

Editing Text

Once you've selected the text you wish to edit, editing can begin. The simplest operation is deletion—to obliterate the text, just press the Delete key (or select "Clear" from the Edit menu, or press the Clear key on the IIGs's numeric keypad).

If you want to replace the selected text with new text, simply begin typing. The selected text will be deleted and the new text you type will be inserted. (Since the entire contents of a line-edit field are selected when you Tab into it, you can just begin typing if you want to replace the default text with new information. You'll get used to filling in and accepting dialogs this way—type something, Tab, type something, Tab, Tab, type something, Return.)

You can also use the Edit menu's "Cut," "Copy," and "Paste" functions on the selection. "Cut" moves the text to the Clipboard and removes it from the text entry area. "Copy" copies the selection to the Clipboard but leaves the original selection intact. "Paste" deletes the selection and replaces it with the contents of the Clipboard. (If no text is selected, the contents of the Clipboard are inserted at the insertion point.)

Simply press the left or right arrow key to deselect the current text without performing any operation on it. If you press the left arrow key, the insertion point will be positioned at the beginning of the selection; if you press the right arrow key, the insertion point moves to the end of the selection. You can, of course, click the mouse to choose a new insertion point as well.

➡ **Shortcut**

If you're filling in a line-edit field and want only to add something to the end—for example, if you're formatting a number of blank disks and want to number them starting at Untitled1 and going up—and the entire field is selected, just press the right arrow key and type your suffix (the disk number in this example) to add it to the end of the default.

Keyboard Translation

II GS typefaces can contain a number of special and foreign characters that you can't type on a normal keyboard. The foreign characters can be typed as follows:

<u>Keystroke</u>	<u>Character Produced</u>
Option-A	Å
Option-C	Ç
Option-E	Places accent on next vowel, e.g. á, é, í, ó, ú
Option-I	Places circumflex over next vowel, e.g. â, ê, î, ô, û
Option-N	Places tilde over next vowel (or N), e.g. ñ
Option-O	Ø
Option-U	Places umlaut over next vowel, e.g. ä, ë, ï, ö, ü
Option-`	Places grave accent over next vowel, e.g. à, è, ì, ò, ù

The System 6 Bonus Pack includes a Desk Accessory (KeyFind) which will help you figure out the other special characters included in most System 6 fonts.

📌 **Note!**

Keyboard translation must be set to "Standard" in the General control panel to access foreign and special characters this way.

More About Windows

Windows are important parts of all Graphical User Interface. (Microsoft even named their entire user interface system Windows!) A window represents a view into a document. A document is any collection of data you're working with—for example, the first chapter of the Great American Novel, or your quarterly report, or the spiffy new drawing you're working on for the cover of the next Weird Al Yankovic album.

A view is just a way of looking at your document. For example, if you're working with a picture in a paint program, the program probably allows you to work with the document at various magnifications—those are your views. Some programs only have one view, but many let you look at your data in more than one format. Since you might not be able to see the whole document on the screen at once (that is, it might be larger than the screen), you can usually scroll the document to view to different parts of it. Windows provide a way of manipulating your views and keeping track of all the documents you have open.

As usual, we'll use a Finder window to illustrate our discussion. In Figure 5L you'll see a couple of windows called "Q1" and "Q2." (They're from one of our Q Drives.) It's not only possible to have more than one window open, it's actually pretty common—the Finder, for example, lets you open twelve windows at once.

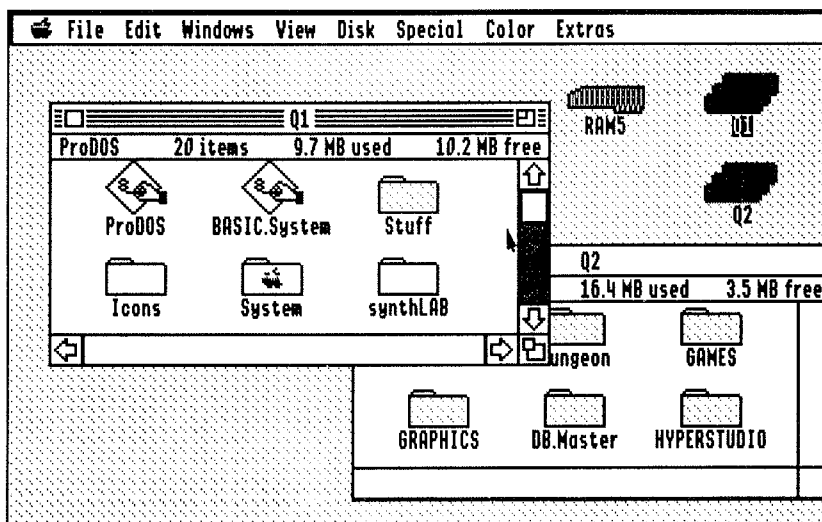


Figure 5L—Sample Windows (from Finder)

▲ **By The Way...**

We've spoken of windows as containing views into documents. It may not be immediately obvious how this applies to the Finder, which actually allows you to directly manipulate your documents in the form of icons. The Finder is a program for manipulating disks and files, so its windows are views into the contents of your disks.

Window Order

If you look closely at Figure 5L, you'll notice that the "Q1" window looks like it's in front of the "Q2" window. And, in fact, the "Q1" window really *is* in front of the "Q2" window. On the Apple Desktop, you work with one window at a time; that window becomes the "frontmost" (or active) window. It's just like having a stack of papers on your desk—you might put the document you're working with on top of the other documents on your desk.

Sometimes the windows don't overlap, so the overlap clue can't always tell you which window is in front. There's another clue, though—the Q1 window also features four black "racing stripes" in its title bar. The title bar is the part of the window that contains the window's name, along with some controls which you'll use to manipulate the window. When you see the stripes, you know exactly which window is ready to go.

To make an inactive window the active window, simply click anywhere in the window. Clicking anywhere in the "Q2" window would bring it to the front and result in the "Q1" window becoming inactive. Some programs, such as the Finder, have a Windows menu that allow you to bring any window to the front easily and quickly—just pull down the menu and choose the window you want to work with.

Opening And Closing Windows

Most programs have an option on the File menu to open documents and to create new documents. Since a window is a view into a document, opening a document opens a window to view the document in. If you choose to open an existing document, you'll first see Standard File Dialog (see Chapter 7) to select the document you want to work on. (In the Finder, you can also double-click an icon to open it.)

When you're done with a document, you can close it. There's usually a Close option on the File menu, but you'll usually find it more convenient to click the close box, the little square at the upper right hand corner of the window.

Moving & Resizing Windows

Desktop windows can be moved anywhere you need them. In our example, it'd be possible to see the complete contents of both windows simultaneously if only we moved the "Q1" window up slightly and moved the "Q2" window down a little bit. That's easy to do via dragging.

Simply position the pointer on the window's title and press the mouse button. While continuing to press the button, drag the mouse. An outline of the window will move with the pointer; position this outline where you want the window to be and release the mouse button. The window will move to the location you choose. It's not necessary that the entire window be visible on the screen—windows can be positioned so that part of the document is off the screen.

 Note!

To cancel the drag, move the pointer to the menu bar and release the mouse button.

To make a window larger or smaller, you'll use the grow box, the small square in the lower right corner of the window. Position the pointer on the grow box and press the mouse button. As you drag the mouse, lines appear on the screen indicating the new size of the window. When you release the mouse button, the window is redrawn at the size you selected.

A faster way to resize the window when you want to view it at its largest possible size is to click the zoom box, the small box at the right end of the title bar. Clicking the zoom box once expands the window to fill the entire screen. Click the zoom box again to return the window to its original size and location.

Scrolling Through Windows

It often happens that the document you're working on is larger than your computer's screen. To overcome this problem, most windows include horizontal and/or vertical scroll bars to allow you to view any portion of your document within the window. In the "Q1" window in our example (Figure 5L), you'll notice two sets of arrows, at the right and at the bottom of the window. The set of arrows on the right of the window allow you to scroll the contents of the window up and down; the set on the bottom of the window allow you to scroll the contents left and right.

- To scroll by a small amount, click the appropriate arrow once.
- To continuously scroll slowly, press the appropriate arrow.
- To scroll by a large amount, click the gray area closest to the arrow that points in the direction you want to scroll. (The gray areas scroll by an entire window-full.)
- To scroll to any arbitrary point within the document, drag the white box between the arrows to a position proportional to the one you want to view.

The scroll bars not only allow you to scroll the window, they also tell you something about our current position within the window. The elevator changes in size according to how much of the document is currently in the window (in our example, about two-thirds of the contents of the window are visible) and its position within the gray area tells us exactly where in the document we are (at the top, in our example). (The elevator never gets so small that you can't grab it—there's a minimum size which is maintained even when the document is very large.)

 Note!

In a word processing document, scrolling the document with the scroll arrows usually does not move the insertion point. Thus, it's possible that the insertion point will move off the screen. Click the I-beam on a new insertion point before you type anything.

Windows With Missing Parts

The Finder windows are what you might call “full-featured windows”—they have all the controls that we’ve discussed here. Windows that can’t be resized won’t have a zoom box or a grow box. Some windows can’t be closed, so they may be missing a close box. Some won’t even have a title in their title bar, or they may be missing a title bar entirely (indicating that they can’t be moved).

And, of course, in some cases the elevator and gray areas may be missing entirely, like the left/right scroll bar in our example—this indicates that the entire document fits horizontally within the window’s current boundaries. If the contents of a window can never be arranged in such a way as to need scrolling in a certain direction, the window might not even have a scroll bar for that direction—some windows always contain the entire document, so they never have scroll bars.

So when you’re confronted with a window, you can tell almost immediately what actions are allowed just by looking for the controls we’ve discussed.

Modal Windows

Modal windows are windows that demand your attention. They have a double-line border all the way around their perimeter and they don’t go away until you’ve dealt with them. You can’t use the pull-down menus or do anything but work the controls in them. We’re talking about alerts and dialogs, of course—they are, in reality, stripped-down windows.

▲ By The Way...

The term “modal” comes from the word “mode.” The Desktop tries to stay as far away from modes as it can—instead of limiting your actions depending on what mode the program is in, you usually can choose to do any operations in any order. Most importantly, you should always have access to the menu bar, instead of having to exit the “working with a document” mode to enter a “menu” mode the way you do in programs like AppleWorks. But sometimes modes just can’t be avoided. When you Initialize a disk in the Finder, for example, it just doesn’t make any sense to let you do anything else until you’ve finished with the dialog.

Windoids

Windoids are, as their name implies, objects that look like windows but aren’t, exactly. The System 6 Video Keyboard (Figure 5M) is a windoid. Like a real window, there’s a “title bar” to allow you to drag the windoid around the screen, but no actual title. The title bar looks undernourished—it’s small and gray. There’s also a close box on the left end of the title bar.

The main difference between windows and windoids is that windoids always stay on top of the window stack. It is impossible to bring a window in front of a windoid. That doesn’t mean that you can’t work as you usually do—windoids don’t actually get in your way. The computer will always work with the topmost real window, ignoring the windoid and behaving as it always did. The only difference is that the windoid is always on your screen and can’t be covered up.

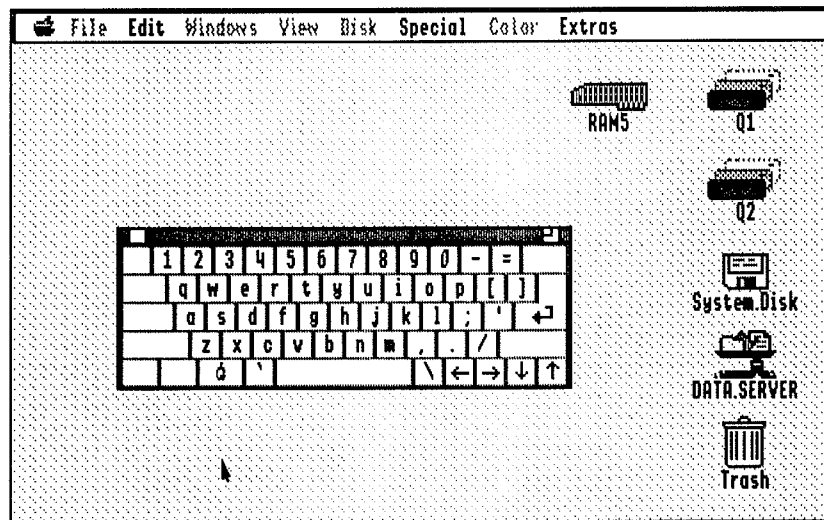


Figure 5M—Sample Windoid (Video Keyboard Desk Accessory)

Programs that let you tear off menus (see the section on menus earlier in this chapter) will turn them into windoids. Clicking an item in the windoid is the same as selecting it from the original pull-down menu. You can, of course, reposition the torn-off menus so that they don't get in your way, and close them when you're done with them.



Welcome To The Finder

The Finder is a special IIGS Desktop program for manipulating your disks, programs, and documents. It's included with the IIGS System Software, and it's the first program most IIGS users learn how to operate. When you boot your System Disk (or the hard drive on which you've installed System 6) and the red bar has finished filling the thermometer, the Finder is normally the first program that appears on your screen.

The Finder has most of the same functions as other Desktop programs, except that these functions work directly on the contents of your disks instead of, for example, on word processing documents. The Finder allows you to:

- Initialize new disks and erase existing ones for reuse
- View your disks' contents and organize files into folders
- Move files from one folder to another
- Copy files from one disk to another
- Rename, lock and unlock, and delete files
- Display information about your disks and files
- Perform routine verification of disks and files
- Launch programs and open documents

We've already seen many of the Finder's menus and dialogs as examples in the previous chapter. And since the Finder is a Desktop program, you only need to learn the features specific to the Finder, not the features all Desktop programs have in common—and you'll find other desktop programs equally easy to learn. Boot up the Finder and follow along.

The Icons

The Finder displays icons (small pictures) to represent the disks and devices you can work with. Figure 6A is a "disktionary" of the disk icons you'll see in the Finder. The icons are arrayed along the right side of the Finder screen, starting at the top. There's also a special icon called the Trash in the lower right corner, which you'll use when you want to get rid of something. Each icon is labeled with a "name tag" to identify the disk. (If any of the disks are write-protected—in other words, if the notch is covered on a 5.25" disk or the window in the corner of a 3.5" disk is open—a small padlock symbol appears to the left of the disk's name.)

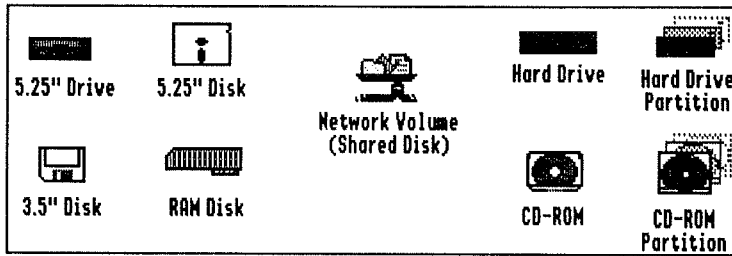


Figure 6A—"Disktionary" of Desktop Icons



Note!

Hard drives, due to a ProDOS limitation that restricts disk sizes to 32 megabytes, may be displayed as multiple icons. Each icon represents one partition, or division, of the drive, and each is accessed as a separate drive, although physically all the partitions are on one drive.



Note!

Due to the nature of 5.25" drives, the Finder can't tell when you insert a 5.25" disk in a drive (or remove one). Therefore, the Finder displays 5.25" drives separately from the disks that are in them. If you have a 5.25" drive with a disk in it, you'll see two icons for it—a drive icon (called APPLEDISK5.25A) and an icon for the disk itself. If you don't have a disk in the drive, you'll only see the drive icon.

If you're following along, your Finder screen should look something like Figure 6B right now—icons on the right side of the screen, Trash in the lower right corner of the screen, and no windows open. (Your system is probably somewhat different from ours, so you probably won't have exactly the same icons we do.) If you have some windows open, close them by pulling down the File menu and choosing "Close All."



By The Way...

The Finder places the icon of disk you booted from in the upper right corner of the desktop, and places the Trash icon in the lower right corner of the screen.

Icons don't always represent disks. The Finder also uses icons to represent files—collections of information stored on disks. You probably don't see any file icons right now, but you will soon enough. "File" is a generic term; every bit of information stored on your disk is stored in a file of some kind.

Files that contain instructions for the computer are called programs. Programs come in two flavors: system software tells the computer how to behave in general terms; applications tell the computer how to perform a specific task—word processing, for example. (The Finder is a difficult program to categorize in this way; it's an application that comes with the system software and is thus often considered to be system software.)

Files that contain information you (or some other human) entered are called documents. The word "document" does not necessarily imply text or printed material; sound and music files, for example, are also considered documents.

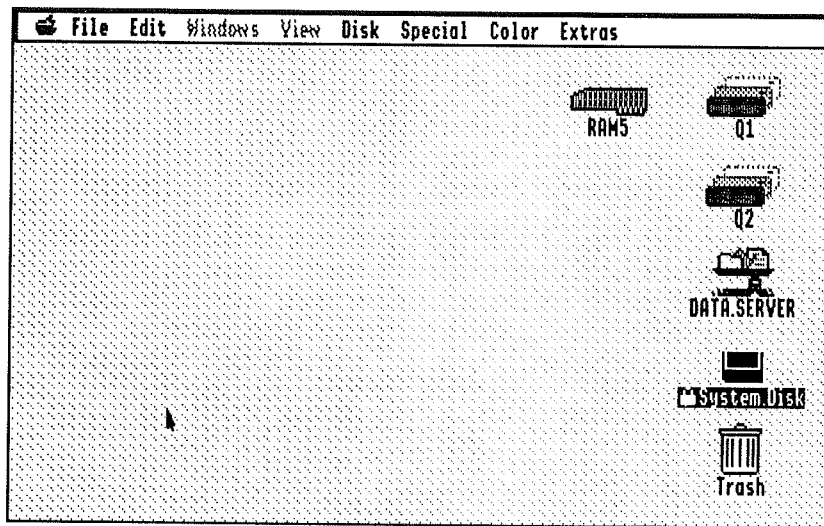


Figure 6B—The Apple II GS Finder

Folders are files that *contain* other files; you can use them to keep the contents of your disks organized. Folders can even contain other folders. Folders are really a special kind of file that the computer uses to keep a list of the things you've placed "inside" the folder, but they're often considered separately from other types of files because of their special properties. When a discussion applies both files and folders, we'll use both terms.

Note!

In the exercises we're about to do—selecting, coloring, renaming, and dragging icons—we'll just be using the disk icons on your desktop (and the Trash) to keep the examples simple. The principles you'll learn, however, apply to all Finder icons, including files and folders.

Working With Icons

Selecting Icons

The Finder, like all Desktop programs, uses the object-action method: first you tell the Finder what object(s) you want to work with, then you tell it what action to perform. Objects (disks, documents, and programs) are represented by icons. Actions are selected from the pull-down menus or performed directly on the icons by manipulating them with the mouse. So before you do anything with the Finder, you tell it which icon you want to do something to.

You select individual icons by clicking them once. Try it—click any icon on the desktop and watch it turn black. Click another. The first icon you clicked returns to its original coloring, and the new icon is selected.

To select more than one icon at the same time, click the first icon, then hold down Shift and click the rest of the icons. The Shift key is a signal to the Finder that it should add the new icon to the group of selected icons instead of making it the only selected icon. You can also shift-click a selected icon to unselect it, should you select one you didn't mean to select. To unselect all icons, click any other icon that's not already selected, or click anywhere that's *not* an icon. Here's a summary of what happens when you click and shift-click in the Finder:

- Click an unselected icon: The clicked icon is selected; all other icons are un-selected.
- Shift-click an unselected icon: The clicked icon is selected; other icons stay selected.
- Shift-click a selected icon: The clicked icon is un-selected; other icons stay selected.
- Click anywhere that's not an icon: All selected icons are un-selected.

You can also select icons by dragging a rectangle around them. To try this out, point to a spot just below and to the right of the Trash. (Don't actually position the pointer on the Trash itself; if you do, you'll end up dragging the Trash around the screen.) Press the mouse button and drag upward and to the left. As you do, a rectangle will appear. One corner is anchored at the point where you started dragging. The corner of the rectangle *diagonally opposite* the fixed corner is attached to the mouse pointer. Drag the mouse until the rectangle surrounds all the icons on your screen, then release the button. All the icons enclosed—or even touched—by the rectangle turn black. If you need to select a group of icons without un-selecting the icons you have already selected, hold down the Shift key while dragging a rectangle around them.

■➡ **Shortcut**

To select all the icons, pull down the Edit menu and choose "Select All" (or press ⌘-A). "Select All" is especially useful when you want to work with most of the icons, but not all of them—just choose "Select All," then shift-click the icons that you don't want.

Opening Icons

Once you've selected an icon (or a group of icons), it's natural to want to see what's inside it. To do this, just pull down the File menu and select "Open" (or press ⌘-O). Since we're working with disk icons (and the Trash, which acts like a disk icon even though it really isn't), opening the icon will produce a window displaying the contents of the disk or the Trash. Try it on the disk in the upper right corner of the screen (see Figure 6C).

■➡ **Shortcut**

To open icons more easily, just double-click them. If you selected a group of icons, double-click any blackened icon to open them all. If you want to open a single icon, you don't need to click it first to select it—the double-click will select the icon and open it. We don't know anyone who ever actually uses the "Open" option on the Finder's File menu.

The window that appears is a standard IIGS Desktop window and can be manipulated like any other Desktop window. If you haven't had a chance to practice moving and resizing windows, you might want to turn back to Chapter 5 and try it out. Otherwise, close the window you opened by clicking its close box or by choosing "Close" from the File menu (or pressing ⌘-W).

As we'll see later, different things happen when you open different kinds of icons. Opening a disk icon the contents of the disk in a window. For now, that's enough—we'll get to opening applications, documents, and folders soon enough.

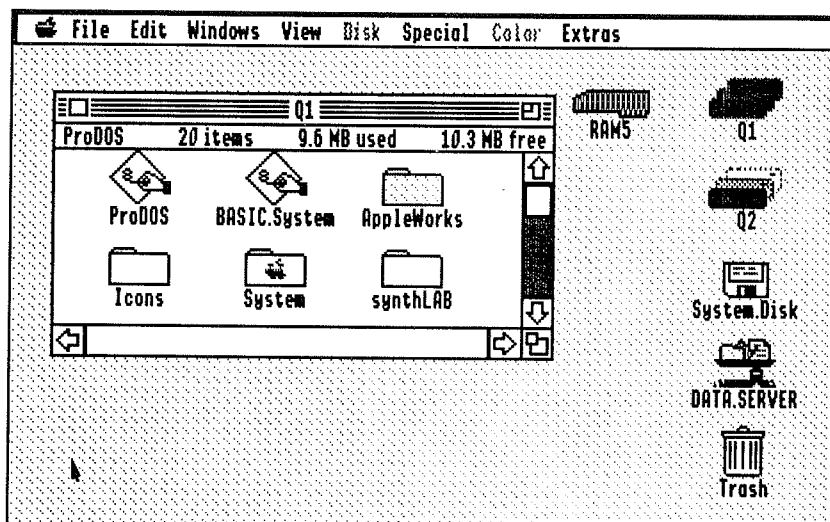


Figure 6C—An Open Window In The Finder

Dragging Icons

You can drag icons around the desktop. Just point to one, hold down the mouse button, move the icon to a new position, and release the mouse button. An outline of the icon follows the arrow around—the icon itself stays where it was until you finish dragging, then it jumps to the new location.

You can also drag groups of icons if you've selected more than one. Unfortunately, the Finder doesn't remember where you put disk icons (or the Trash), so even if you move the icons to the other side of the screen, they'll be back on the right side the next time you return to the Finder. (The Finder does remember the positions of *files* you put on the desktop, but we haven't talked about that yet.)

More interesting is what happens when you drag one icon *onto* another. If you don't have a 3.5" disk icon on the desktop, put your System Disk in a drive. (If you already have a 3.5" disk icon on the desktop, we'll use that one for this experiment. If you have more than one, it doesn't matter which one you use.)

Drag your 3.5" disk icon to the Trash. More specifically, position the pointer on top of the 3.5" disk icon and press the mouse button. Hold the mouse button while you drag the mouse to the Trash icon—an outline of the 3.5" disk icon will follow the pointer. Position the pointer on top of the Trash icon. Make sure the Trash icon has turned black (if it hasn't, your pointer isn't in the right place) and release the mouse button.

If you did things right, your 3.5" disk drive whirled and spat out the disk. The disk should now be gone from the desktop. (If you ejected the disk you started up from, a "shadow" of the disk will remain on the desktop—you can't permanently eject your startup disk.) For now, just put the disk back in the drive and the 3.5" disk icon will reappear.

Other disk icons can be dragged to the Trash, too. For example, if you're on an AppleShare network, you can drag a shared volume icon to the Trash to unmount it. You can also use the Trash to make the Finder forget about the disk in a 5.25" drive—just drag the 5.25" disk icon to the trash. You can't put hard drives, RAM Disks, or most other disks in the Trash—only disks with removable media and network disks can be "ejected" or "unmounted" this way.

The Trash is used for removing icons from the Finder desktop. In the case of disks, as we've seen, this means ejecting them or unmounting them. Dragging files and folders to the trash, on the other hand, actually removes them from the disk they're on. We'll get to that later, though.

NB

Coloring Icons

The Finder also allows you to color your black-and-white icons (such as the Trash). For this example, we'll color the Trash green, like one of those big industrial trash barrels. Just click the icon, and then pull down the color menu. Choose the medium green (in the middle of the third column) and release the mouse button. The trash can will instantly turn green. In effect, the Color menu changes the *white* in the icon to whatever color you choose.

You can also color an icon's outline (the black lines in the icon). Let's try that. Click the Trash again. Now press the Control key and hold it down while you select the desired color from the menu—let's try the yellow at the bottom of the second column. Release the mouse button and your Trash icon appears green with yellow highlights.

Note!

When you color an icon or its outline, the Finder automatically unselects it. (If it remained black, you wouldn't be able to see the change you made.)

The Finder cannot color icons that already contain colors. For example, you can't color your hard drive icons, since they contain gray and red; you can't color RAM Disk icons, since they contain green and yellow; you can't even color 3.5" disk icons, since they contain gray. In fact, the only icon on a normal desktop that you *can* color is the Trash. You can, however, color most application, document, and folder icons, which is a tremendous aid in keeping your disks organized and easy to use.

 Note!

The Finder does not remember the colors you've assigned to the Trash. So our little exercise in coloring the Trash was, well, pretty pointless—but the Finder does remember the colors of application, document, and folder icons.

Renaming Icons

Renaming an icon is as simple as clicking the icon and typing the new name. If the icon you want to rename is already selected and has been used in another operation, or if more than one icon is selected, you will need to unselect the icon (click the desktop or another icon) and click the icon you want to rename once, *then* type the new name. (Or just press Return first.) Press Return when you're finished, or, if you decide not to rename the icon after all, click the icon once or press Escape to restore its original name.

 Note!

If you're changing the name of a ProDOS disk (or a file on a ProDOS disk), remember that ProDOS names must begin with a letter, can be up to fifteen characters long, and can include letters, digits, and periods. If you're changing the name of a Macintosh (HFS) disk, the name may be up to 27 characters long and contain any character but a colon (:). Names of files and folders on HFS disks (and network volumes) may be up to 31 characters long.

 Shortcut

If you want to make only minor changes to an icon's name, you can treat the icon's name as a miniature line-edit field. Position the mouse cursor over it; when the I-beam cursor appears, use the mouse to select a section of the name for editing. Or use the arrow keys to edit the name. See the last chapter's section on line-edit fields for more tips on editing.

System 6 does not allow you to have two disks with the same name in drives at the same time. Since the IIGS identifies your disks by name, it would be confusing—to you *and* to the computer—if you had two disks with the same name on the desktop. Therefore, you can't change a disk's name to be the same as a disk that's already on the desktop. (You must first remove one of the disks from the desktop.) You also can't change the name of the Trash or of your 5.25" drives (you can change the name of 5.25" *disks*, though). Curiously, you *can* name a disk "Trash," though we suggest that you don't.

 Note!

System 6 disregards upper and lower case differences in disk and file names. For example, disks named "System.Disk," "system.disk," "SYSTEM.DISK," and even "SyStEm.dIsK" all have the same name as far as the computer's concerned.

Copying Disks

Making backup copies of your disks is simple. Just drag one disk onto another icon of the same type (for example, 3.5" disk to 3.5" disk). The Finder will ask you (via an alert) if it's OK to erase the contents of the disk you're copying to (the target disk) and replace them with the contents of the original disk (the source disk). If you click the "Replace" button (or press Return), the copy operation will begin. Since the two disks are the same size, an image (or disk-to-disk) copy is performed, making an exact copy of the disk very quickly.

If the two disks are of *different* sizes, the Finder will create a folder on the target disk (giving it the same name as the source disk) and copy the contents of the source disk into that folder. The copy proceeds one file at a time and is thus known as a file-by-file copy. No information is erased from the target disk; the contents of the source disk are *added* to the existing contents of the target disk. This makes it easy to copy a new program to a hard drive, where each program should be in a separate folder—in fact, it's virtually automatic.

Note!

Copying a disk file-by-file works best if the disk you're copying is smaller than the disk you're copying to. If you drag a larger disk onto a smaller disk, you probably won't have enough room on the target disk for all the contents of the source disk, but the Finder will attempt the copy operation anyway. (Your data won't be damaged; the Finder will merely cancel the copy when it figures out there's not enough room on the target disk to do your bidding.) And remember—3.5" disks hold more data (800K) than 5.25" disks (140K) even though they're smaller.

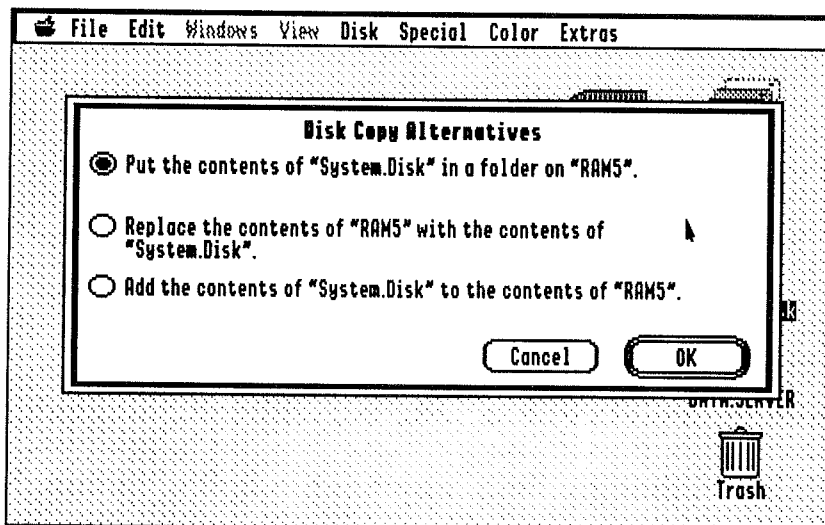


Figure 6D—Disk Copy Alternatives

Note!

Hold down the Option key while dragging one disk onto another for a dialog of Disk Copy Alternatives (Figure 6D). If you choose the first option, "Put the contents in a folder," the Finder will do a file-by-file copy, as if you were dragging a disk onto another disk of a different size. If you choose the second option, "Replace the contents," the Finder will attempt a disk-to-disk copy. If you choose "Add the contents," the Finder does a file-by-file copy, but merges the contents of the two disks instead of making a folder for the contents of the new disk.

The "Copy in progress" dialog (see Figure 6E) contains a thermometer, a red bar which indicates how much of the operation has been completed. The number in the box tells you how many blocks or files remain to be copied, and a status message lets you know exactly what the Finder is doing at each stage of the copy operation.

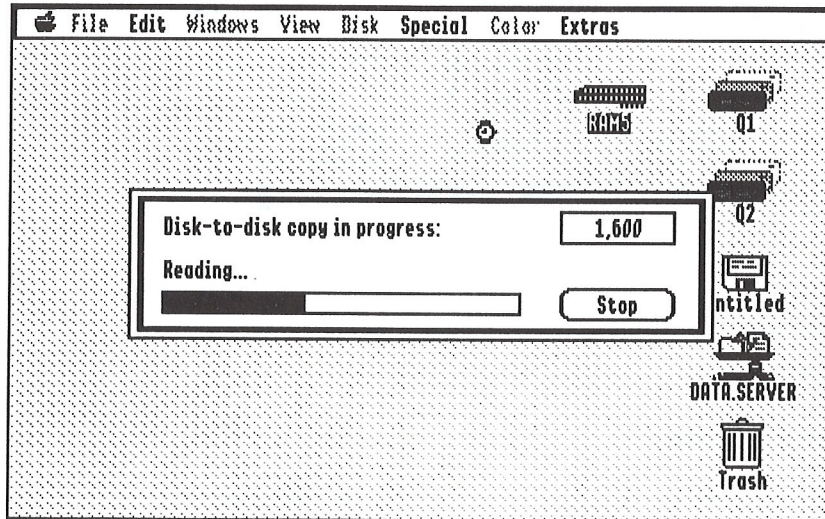


Figure 6E—Copy In Progress Dialog

There's also a Stop button. During a disk-to-disk copy, the Stop button stops the copy with no loss of data *only if the Finder hasn't written any data to the target disk*. If the Finder *has* written to the target disk, the Finder naturally can't *unwrite* it. During a file-by-file copy, any files that were written to the target will be left in place if you Stop the copy; if you decide you don't want the files there, you will have to remove them yourself. (See the last chapter's section on buttons for more information on the differences between the terms "Cancel" and "Stop.")

Warning!

If you initiate a disk-to-disk copy onto a disk you didn't mean to overwrite, then realize your mistake and hit Stop, the original information may be partially or totally erased. Be completely sure you want to overwrite the contents of the target disk before clicking "Replace."

If, during a file-by-file copy, the Finder discovers that a file already exists on the target disk with the same name as a file being copied, it will display an alert (Figure 6F) to find out how to handle the duplicate file. You can choose to Skip or Replace this item only (the Finder will display another alert when it encounters another duplicate file), to Skip or Replace all duplicates the Finder encounters during the operation, or to Cancel the entire copy operation.

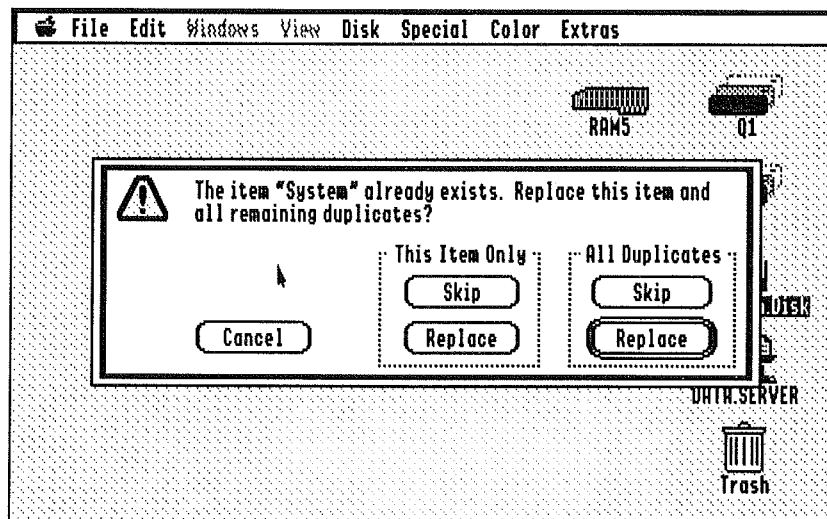


Figure 6F—Replace/Skip Duplicate Files Dialog

Initializing & Erasing Disks

Initializing and erasing disks are two very similar operations the Finder can perform on disks. Both have the effect of completely erasing every bit of information on the disk. Usually you *initialize* blank disks that have never been used; you *erase* disks that have already been initialized but have data on them that you no longer need. The procedures for the two operations are nearly identical, so we'll cover them together.

Warning!

Remember, both "Initialize" and "Erase" will totally erase everything on the disk. In the case of "Erase" some data recovery may be possible with programs like ProSel-16 or Salvation-Deliverance, but usually your data will be gone for good. So be careful!

Warning!

The Finder will automatically attempt to format blank 3.5" disks inserted in a 3.5" drive. If you insert a disk which is supposed to have information on it and the Finder asks you if you want to initialize it, the disk is probably damaged. Unless you really do want to initialize the disk, we suggest clicking the "Eject" button so that any remaining data can be salvaged by a disk repair utility such as ProSel-16 or Salvation-Deliverance.

⑥ New & Improved

System 6 can read, write, and initialize disks in HFS (Macintosh) format as well as in ProDOS (Apple II) format if the HFS File System Translator is installed. System 6 can also read Apple DOS 3.3 and Apple Pascal 5.25' disks with the appropriate File System Translators installed, but it can't write or initialize disks in these formats (the FSTs are read-only)

To perform an initialize or erase operation, first click on the disk you want to erase, then pull down the Disk menu and select "Initialize..." or "Erase...". The dialog which appears (Figure 6G) allows you to specify the new name of the disk (the default entry for the disk name is the disk's current name, or "Untitled" if the disk is blank).

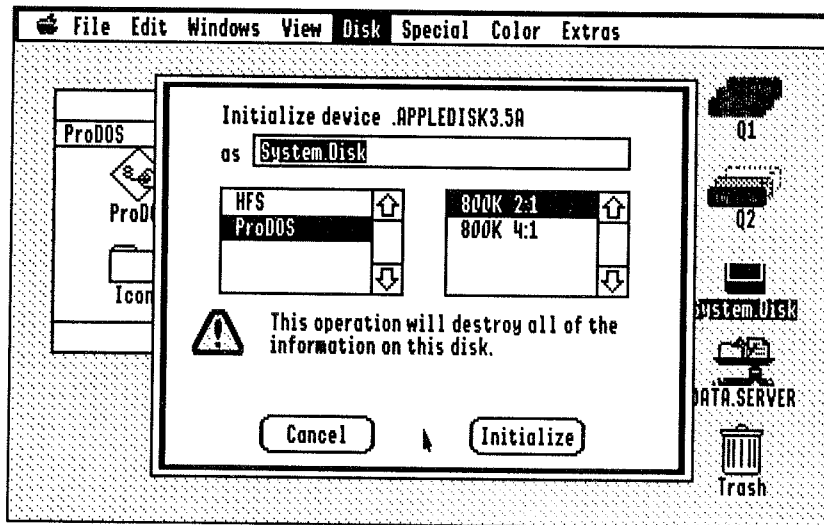


Figure 6G—Disk Initialization Dialog

If you're making a ProDOS disk, keep in mind that ProDOS disk names must begin with a letter and that they can be up to fifteen characters long and can include letters, digits, and periods. Macintosh disk names can be up to twenty-seven characters long and can contain any character except a colon. Don't choose a name that's already used by a disk on the Finder desktop.

Next, select the desired format of the new disk from the list on the left. If you're making a new 5.25" disk, your only choice is ProDOS, since the Macintosh does not support 5.25" disks. If you're using a 3.5" drive, and will be using the disk only on the IIGS, or if you will need to use it with older programs like AppleWorks, choose ProDOS. If you will be using the 3.5" disk to transfer files back and forth between the IIGS and the Macintosh, choose HFS.

Note!

HFS We don't recommend formatting your hard drive partitions or RAM Disk with HFS because you won't be able to access these drives from older (ProDOS 8) programs and because access will be slower than ProDOS format. We suggest initializing only 3.5" disks in HFS format, and only if they will be used to exchange data with Macintosh programs.

Note!

If you're making a new startup disk for your computer, it must be initialized in ProDOS format. You can't start up the computer from a HFS disk.

4:1 = 11C Finally, select the interleave (initialization operations only—you don't have this choice when erasing a disk). If you're making a new HFS disk which will be used primarily on the Macintosh, select 2:1. If you're making a new HFS disk and will be using it heavily on the IIGS, select 4:1. If you're creating a ProDOS disk, select 2:1 except in the rare circumstance that you have (or will be using the disk on) an Apple UniDisk 3.5 drive (the *white* Apple 3.5" drive usually used on the Apple IIc)—in that case, select 4:1. (The interleave option adjusts the disk's formatting so that it can be accessed most rapidly on the machine it will be used on most of the time.)

Note!

If you're unsure which format or interleave or disk format to choose, just accept the dialog as it is. The default settings will work for most users.

By The Way...

You can't change a disk's interleave without reformatting it. If you need to change a disk's interleave for better performance on another system, format a new disk with the desired interleave and copy the old disk's contents to the new disk.

Initialization (or erasure) will begin when you click the "Initialize" or "Erase" button at the bottom of the dialog (or press ⌘-Return). Erasing a disk takes only a few seconds; initializing can take a minute or so. When the operation is complete, the new disk will appear on the desktop.

By The Way...

It is sometimes possible to recover data from disks that have been accidentally erased (not initialized) using utilities like ProSel-16 or Salvation-Deliverance. Don't count on it, though—always make sure you don't need any of the information on a disk before erasing it.

Note!

We suggest using the Initialize option when reusing older disks. The Initialize option will rewrite and verify every part of the disk to make sure it's still good. Additionally, initialization makes sure that the interleave on the disk is set for maximum access efficiency.

Verifying Disks

Disk It's often useful to verify a disk to make sure it's still readable—for example, when you haven't used a disk for a long time or when you're having any kind of trouble accessing the disk. The Verify operation reads all the blocks on the disk to ensure that they're all readable. Just click the icon you want to verify and select "Verify" from the Disk menu. If any blocks can't be read, the Finder will display the bad block numbers in a window.

File This low-level verification does not check the integrity of the files on the disk, nor does it ensure that the data contained in the disk is valid. (It's possible for each block on the disk to be perfectly readable, yet for the disk itself to be unusable due to corrupted folder data or other bad information stored on it.) To verify that each *file* on the disk can be read, click the disk icon and select "Validate" from the File menu. If any files can't be read, the Finder will display the names of the bad files in a window.

If either of these tests detects a problem on your disk, copy any readable files to another disk and stop using the bad disk immediately. Programs such as ProSel-16, Salvation-Deliverance, and RepairWorks can often retrieve some or all of the information from defective disks.

Note!

You can't Verify network volumes, since individual disk blocks are not available across an AppleShare network. You can, however, Validate network volumes.

Working With 3.5" Disks

We've already talked about one way to eject a 3.5" disk—by dragging its icon to the Trash. You can also eject a 3.5" disk by clicking its icon and selecting Put Away from the File menu, or by pressing ⌘-Y. This permanently ejects a 3.5" disk—the Finder will forget about it and stop displaying an icon. (This doesn't mean that you can't put the disk back in, only that the Finder won't keep track of it.)

To temporarily eject a 3.5" disk, simply press the eject button on the front of the drive, or click the disk's icon and select "Eject" from the Disk menu (or press ⌘-E). When you temporarily eject a disk, the Finder displays the disk as a *dimmed* icon, reminding you that it's not in any drive. If you try to work with that disk (for example, copy another disk or file onto it), the Finder will remind you to put the disk back into a drive.

This latter is the key to copying disks if you have a single 3.5" drive. You'll usually put the target disk in first, then eject it temporarily. Then you'll insert the source disk and drag it onto the dimmed icon of the target disk. The Finder will tell you when to switch disks and will ask for the disk it needs by name.

⑥ **New & Improved**

System 6 automatically detects 3.5" disk insertions when it requests a particular disk. There's no need to click "OK" or press Return when the computer asks for a 3.5" disk; just put the disk in any available drive.

To make the Finder forget about a temporarily ejected disk, drag the dimmed icon to the Trash (or select “Put Away” from the File menu). It’s not necessary to insert the disk before getting rid of the dimmed icon.

Working With 5.25” Disks

Because the Finder can’t tell when you remove and insert 5.25” disks., it displays both the drive *and* the disk as icons on the desktop. Anytime you remove a 5.25” disk and insert a new one, double-click the drive icon to tell the Finder that you’ve switched disks. The Finder considers the disk you took out temporarily ejected (showing a dimmed icon) and adds the new disk to the desktop. The Finder will ask you to insert the removed disk again if you try to do anything with it—just insert the disk and click the “OK” button (or press Return).

To permanently eject a 5.25” disk, click the disk’s icon and select “Put Away” from the File menu (or drag the icon to the Trash). Then insert the new disk and double-click the 5.25” drive icon. The old disk’s icon will be completely removed and the new disk’s icon will appear. (You can also do it in the reverse order—drag the dimmed icon to the Trash after the new disk appears on the desktop.) If you don’t want to put a new disk in the drive, just drag the disk’s icon to the Trash and remove the disk from the drive.

If you insert a blank disk in a 5.25” drive and double-click the 5.25” drive icon, the Finder will grind the drive for a while, then ask if you want to initialize the disk. (The same thing may happen if you double-click the icon when no disk is in the drive.) We’ve already discussed the Initialize dialog earlier in this chapter.

▲ By The Way...

The Finder automatically checks your 5.25” drives for disks when it starts up. If you don’t use 5.25” disks very frequently, the grinding sound your empty drive makes when the Finder tries to read it will eventually begin to annoy you. When it does, turn off the “When starting up, check 5.25” drives” check box in the Finder Preferences (accessed under the Special menu). The Finder will then display your 5.25” drive icons but won’t automatically display the disks in them. If you want to use a 5.25” disk, just insert it and double-click the drive icon.

⑥ New & Improved

The Finder preference for not looking at 5.25” drives unless you double-click them is a new (and welcome) feature of System 6.

Working With Hard Drives & CD-ROMs

As we mentioned earlier, your hard drive may be divided into multiple partitions, each with its own icon. The ProDOS format only supports drives 32 megabytes and smaller, so drives larger than 32 megabytes must be partitioned into two or more partitions, each smaller than 32 megabytes. To the IIGS, each partition is a separate disk, and you’ll manipulate them as if they really were separate, even though they’re all physically on the same drive.

▲ By The Way...

You can get around this limit by re-initializing part of your hard drive in HFS (Macintosh) format. (You can't reformat the whole drive for HFS; System 6 must start up from a ProDOS partition.) However, access to HFS partitions is slower than access to ProDOS partitions, and HFS partitions are inaccessible to older (ProDOS 8) programs like AppleWorks. Our suggestion is to stick to ProDOS unless you're swapping the drive between a IIGS and a Macintosh.

HFS
HARD
DRIVE

Most hard drives can't be "ejected," so the "Eject" and "Put Away" menu options have little effect on them. But some hard drives do have cartridges that can be ejected—45 megabyte drives based on the SyQuest mechanism are the most popular. Ejecting a cartridge manually (using the controls on the drive) ejects it temporarily; the Finder displays a dimmed hard drive icon like other ejected disk icons. (You can use this feature to copy information from one cartridge to another—it's cumbersome but it works.) To eject a cartridge permanently, drag the dimmed icon to the Trash (or do a "Put Away"). If there's more than one partition on the cartridge, you'll need to drag *all* the icons to the Trash.

CD-ROM (Compact Disc) drives behave similarly to cartridge drives. However, many CD-ROMs have both Macintosh (HFS) and High Sierra partitions in addition to the usual ProDOS partitions. (High Sierra is a universal format designed to allow CD-ROM discs to be used on many different models of computers.) All these types of partitions will appear on your desktop. As with cartridge drives, the Finder considers the disk temporarily ejected when you remove the CD-ROM. Drag the dimmed partitions to the Trash to permanently eject them.

Working With Network Volumes

If you're on an AppleShare network, you'll probably have shared disks (or network volumes) on your Finder desktop. The process of telling the IIGS which network volumes you want to use is called logging on to the AppleShare file server, or mounting the network volumes. It's accomplished from the AppleShare control panel.

To unmount a network volume, just drag it to the trash, or Eject it or do a Put Away. (Eject and Put Away have identical effects on network volumes.)

Note!

Since all AppleShare servers are Macintosh computers, network volumes follow the same naming rules as Macintosh disks. Disk names can be up to 27 characters in length and can contain any character but colons; file and folder names also cannot contain colons and can be up to 31 characters long.

As we mentioned earlier in this chapter, all the information you put on a disk is stored in files. The System Software itself is stored in a collection of files, and every program you use and every document you create is also stored in a file. The word “file” is a generic term for any collection of data. Let’s take a look at some of the files on your System Disk. Put your 3.5” System Disk in a drive and double-click it.

A window appears displaying the contents of the disk. The window, which can be moved, resized, and scrolled like any standard IIGS window, contains four icons—two folders called System and Icons, and two program files called ProDOS and BASIC.System. Just below the title bar is the info bar or the header—it tells you what kind of disk this window is from (ProDOS, in this case; other possibilities are HFS, DOS 3.3, Pascal, High Sierra, and AppleShare), how many files are in the window, how much space is used by the files on the disk, and how much space is left on the disk.

A disk’s icon and its window are intimately linked. In fact, in many ways, a disk’s icon and its window are *the same thing*. If you drag a file onto a disk’s icon, it appears in the disk’s window. You can also drag a file directly into a window to copy it to a disk. If you drag a file out of a disk’s window and onto the same disk’s icon, nothing happens, because you’ve told the Finder to move the file to where it already is.

The concept of each window being linked to a particular icon is an important one. (To find out which icon a given window is linked to, just click the white space inside the window. The corresponding disk icon will turn black. It’s now selected for any operation you care to perform on it—just as if you’d clicked the icon directly.)

The two folder icons represent, logically enough, folders—the two diamond-shaped icons with hands, which resemble sheets of paper being written on, are programs. (There’s an “8” written on the icons, indicating that they’re 8-bit programs.)

▲ By The Way...

Many programs have custom icons. If you see an icon you don’t recognize, it’s usually possible to make a guess. Icons with folded-down corners are usually documents; icons that resemble folders usually are; diamond-shaped icons are usually applications. When in doubt, use the “Icon Info” option on the Special menu to find out exactly what an icon represents.

Icons in a window can be colored, moved around, and renamed, just like icons on a desktop. (You can’t have two icons in a window with the same name, just as you can’t have two disks on the desktop with the same name.) The Finder remembers the colors you assign to icons in windows, as well as the positions of the icons within the window and the position (and size) of the window on the screen.

Note!

Your Desktop is saved when you close a window, when you eject a disk using "Put Away," when you shut down or quit the Finder, and when you launch a program or open a document.

Remember, you can only color black and white icons. You might assume, then, that you can't color folder icons, since they're yellow. Actually, you *can* color folder icons—they're an exception. (Folders are actually white to begin with—the Finder just colors them yellow the instant they're created. They really are black and white icons at heart.) You can also color the other icons you see in the System Disk window.

You'll find the "Clean Up" option (on the Special menu) useful for keeping your icons organized. It moves all your icons so they're evenly spaced along an invisible grid. Try dragging some of the icons in the window a short distance, then choosing "Clean Up" and watching the icons dance back to the nearest point on the invisible grid. (Think of the "grid" as a bunch of evenly spaced "magnets" beneath the window's surface that "pull" the icons to them.)

Changing Views

The View menu is useful for displaying the contents of a window in different formats. "By Icon" is the default view; there's also a "By Small Icon" which allows you to see more files in the same window space. Instead of showing the icon's name *under* the icon, the name is displayed to the *right* of a miniature icon. When you first switch to the "By Small Icon" view, the icons will have a lot of space between them; you can move them as desired and choose "Clean Up" to align them.

Shortcut

A faster way to arrange your "By Icon" or "By Small Icon" view is to hold down the Option key while you pull down the Special menu. The "Clean Up" item will now read "Clean Up By Name." When you select it, the Finder will pick up all your icons and rearrange them alphabetically in the grid. If you're viewing by icon, "Clean Up By Name" arranges the icons into rows so that you can scroll the window up and down to view them. If you're viewing by small icon, the Finder arranges the icons in columns so that you can scroll the window left and right to view them. "Clean Up By Name" doesn't permanently alphabetize your icons—new icons will be added to the bottom of the window—but it's a good place to start.

ICONS
IN
WINDOWS

The other views—"By Name," "By Date," "By Size," and "By Kind"—are list views, not icon views. In list views, you can't drag the icons around to new positions; positions of the icons are defined by the view you've selected. You can't drag a rectangle around a group of files to select them; you must instead use the shift-click technique to choose multiple files. List views can only be scrolled vertically, not horizontally; resize the window to read the other columns of information (size, kind, and modification date). The list views are:

- *By name:* Lists files alphabetically by name
- *By date:* Lists files according to the date they were last changed, most recent first
- *By size:* Lists files according to the amount of space they use on the disk, largest first
- *By kind:* Lists files according to the kind of file they are—similar files (for example, AppleWorks documents) are grouped together alphabetically

Folders

We've mentioned that folders are special kinds of files that can hold other files. Let's look inside a folder on the System Disk. Double-click the System folder icon.

Imagine that. *Another* window! This window is titled System and is linked to the System folder icon in the System.Disk window. The new window is probably blocking your view of most of the System.Disk window, so move it out of the way by grabbing the title bar with the mouse pointer and dragging it to an unused portion of the screen.

The System window is linked to the System folder icon in just the same way the System.Disk window is linked to the System.Disk icon. They are, for all practical purposes, the same thing in two different guises. When you're looking at the window titled System, it's just as appropriate to say that you're looking at the System folder, because the window *is* the folder.

We say that the System folder's parent is the System.Disk, because the folder icon is in the System.Disk window. (The System window is also a child of the System.Disk window.) A given folder has only one parent (because it can only be inside one other folder or disk) but it can have many children (because you can put as many folders as you want in a disk or folder.)

In the System window is another set of icons, some of them folders. Browse through them using the scroll bar on the right side of the window.

The idea of folders is an important concept, not only on the Apple IIGS but on most other computers you'll encounter. Imagine trying to find anything on a hard drive where everything is displayed in one window! The System Disk itself contains over 40 files and folders; your hard drive includes all these files and more if you've installed System 6 on it, plus files for all the programs you've installed and your documents. It wouldn't be *too* bad on a floppy disk (even a 3.5" disk), but even viewing files by name or by type won't help you keep things organized when several megabytes' worth of files are all displayed in one window.

That's what folders are for. The System files are organized into folders by their function. For example, fonts (typefaces) go in one folder; desk accessories go in another; sounds go in a third. All of these folders (plus a few others) are stored inside the System folder. So when you want to find the system files, you know just where they are—inside the System folder and the various folders it contains—but you're not faced with all those files every time you open a disk. The system files are grouped together and assigned a name, via the concept of a folder.

You can create your own folders, too. Just use the "New Folder" option on the File menu (or press ⌘-N). A new folder called "Untitled" will appear in the frontmost window. (You may need to scroll the window to find the new folder if the window is crowded.) The folder is selected and ready to rename—just type the name you want to give the folder and press Return.

Most Finder users create one folder on their hard drive for each of their programs, then place each program's files in the folder assigned to it. Additional folders are created to store documents used with these programs. With folders, you can keep yourself organized and avoid spending your valuable time hunting for that one important document.

Window Management

Now let's open a few more windows—double-click the Fonts and Desk.Accs folders. We're not really concerned about the contents of these windows right now; we'll be using them to illustrate a few handy window management techniques. The screen tends to get cluttered when more than a couple of windows are open; these tricks can help minimize the confusion.

You're already familiar with the ways you can manipulate IIGS Desktop windows like those you see in the Finder. You can drag them around, resize them by dragging their zoom box, and you can bring any window to the front by clicking anywhere in it. You can close a window by clicking its close box, by choosing "Close" from the File menu, or by pressing ⌘-W.

You can also close *all* open windows by choosing "Close All" from the File menu. Even experts get confused sometimes, and closing all open windows returns you to a clean Finder slate.

You may also find it useful occasionally to close other open windows as you open new ones. There's nothing at all that says you have to keep the System.Disk and System windows open once you get to the Fonts window. The trick we'll discuss in a minute gives you a way to reopen a window's parent folders without having to go all the way back to the original disk icon.

➡ **Shortcut**

You can also close all open windows by holding down the Option key while you close a window, or while you press ⌘-W. You can drag a window without bringing it to the front of the stack by holding ⌘ as you click in its title bar.

▲ **By The Way...**

We're not quite sure what ⌘-W is supposed to be short for—maybe "window"—but perhaps it was chosen because it's on the left side of the keyboard. It's very easy to zip through Finder windows with your left hand on the keyboard and your right hand on the mouse. The ⌘ key is right there, along with the Shift key for shift-clicking, A for Select All, E for Eject, and W for Close. It's not really a shortcut, just a useful observation.

Another big help is the Window menu. When the window you want to work with is buried under two or three other windows, just pull down the Window menu for a list of all open windows, then select the one you want to work with. No need to close or move the other windows. Or choose Stack Windows from the Window menu; all the windows will arrange themselves neatly on the screen, ready for you to choose one. The order of the windows doesn't change, only their positions. (Open up some windows and try that one out. It's nifty.)

➡ **Shortcut**

Holding down Option while selecting a window from the Windows menu closes it without first bringing it to the front.

The Pop-Up Folder Menu

Occasionally you'll open two windows with the same name. For example, it's possible to open the System folder from two different disks if you want to copy files from one disk to another—it's even possible to have two folders with the same name on the same disk, as long as they have different parent folders. You might even have a disk named *System* *and* a System folder on your hard drive.

So once you've opened two windows called *Fonts*, for example, how do you keep track of which one is which? Simple—hold down the ⌘ key while you press the menu title. A pop-up menu will appear under the pointer, listing the folder itself and, under it, all the folders that the folder is in, all the way back to the disk name (see Figure 6H). (This trick only works on the front window—holding down ⌘ while clicking a background window is how you drag a window without bringing it to the front of the stack.)

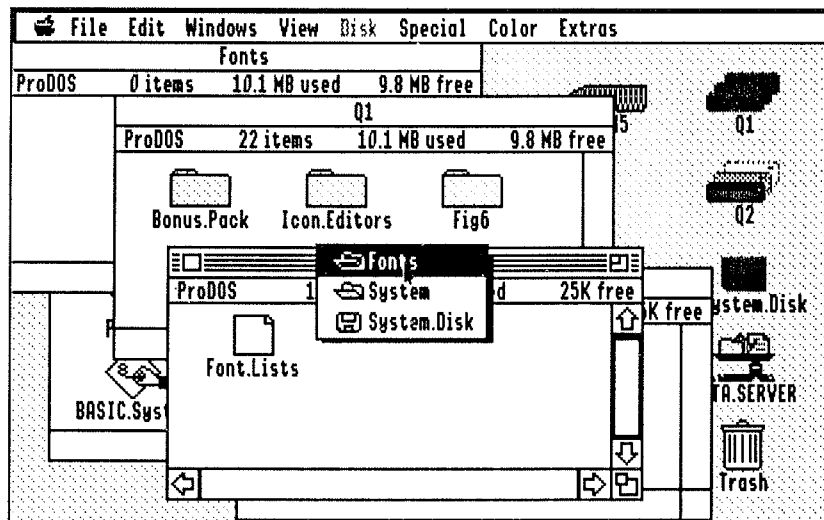


Figure 6H—Pop-Up Folder Menu

In this case, the pop-up menu lets us know that this is the *Fonts* folder, which is inside the *System* folder (the *Fonts* folder's parent), which is on the disk called *System.Disk* (the *System* folder's parent). If you just wanted to know which window belonged to which disk, now you know, so you can release the mouse button. However, it *is* a pop-up menu—you *can* select a folder or disk name from the menu. If you do that, the Finder will open the folder or disk (or bring its window to the front if it's already open).

⑥ New & Improved

The pop-up folder menu is a new feature of System 6.

Tunneling

It sounds like something a rodent might do (appropriate on a mouse-driven computer system), but tunneling is really just a handy way to automatically close old windows as you open new ones. Follow along on a simple demonstration.

Close all the open windows except for the main System.Disk window. Now hold down the Option key while you double-click the System folder. As usual, the System window appears—but then the System.Disk window disappears. Congratulations, you have just tunneled!

Using tunneling, you can go deep within your disks' folder structure without ever becoming overwhelmed by the number of open windows on your screen. Try it again—tunnel into the Desk.Accs folder. You're now three "levels" into the System Disk with only one open window!

If you want to go backward along the folder path, the pop-up folder menu comes to the rescue. Hold down the ⌘ and Option keys while you select an item from the pop-up menu, and the current folder will disappear after the new one is opened. This feature is called reverse tunneling. Now you can move up and down through the various "levels" of your disks while keeping only one window open from each disk. As you can imagine, tunneling can considerably ease the confusion caused by a dozen open windows.

⑥ New & Improved

Tunneling and reverse tunneling are new features of System 6.

Windows, Icons, and Directories

When we're working with icons and windows in the Finder, we're really working with directories. A directory is a special file which the computer uses to keep track of all the *other* files on the disk. Each disk has a volume directory, or main directory, that is used to keep track of the files you see when you open the disk. Folders are stored as subdirectories, which are secondary lists of files that appear in the main directory. Subdirectories can, of course, contain other subdirectories.

Note!

ProDOS disks can contain only 51 files in their main directories. Some of these files can be subdirectories, of course, and subdirectories can contain a virtually unlimited number of files. (Under ProDOS, folders are not only useful for organizing your files, they're a necessity for getting around the 51-file limit.). Macintosh (HFS) disks have none of these limitations.

Both the main directory and subdirectories can be generically called directories. The Finder displays the contents of all directories graphically in windows. Main directories are represented by disk icons; subdirectories are represented by folder icons.

In some programs, you'll be asked to specify pathnames for a file. A pathname is just a list of every folder the file you're accessing is in, starting with the name of the disk it's on and ending with the name of the file it's on. For example, the pathname of the Calculator desk accessory, if you've installed it on a Q Drive, is :Q1:System:Desk.Accs:Calculator. This tells the computer that the file is on the disk Q1, and that to get to it you must first open the System folder and then the Desk.Accs folder before finding the Calculator desk accessory (Figure 6J).

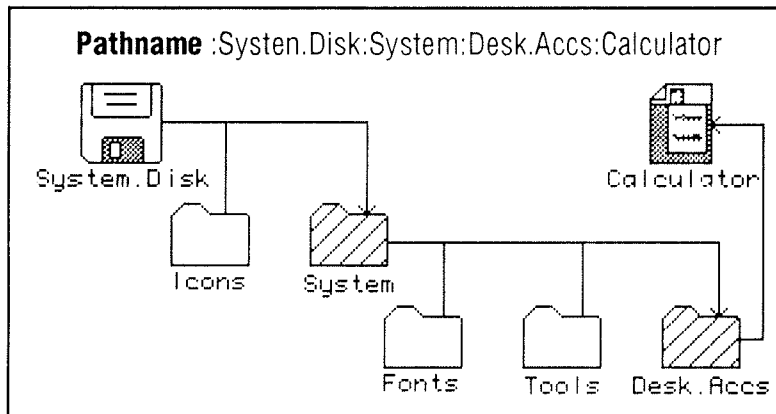


Figure 6J—How System 6 Follows Pathnames

▲ By The Way...

In System 6, you can use colons (:) or slashes (/) to separate the folders in a pathname, but you have to be consistent (you can't use both separators in the same pathname). When you're using a ProDOS 8 program, like AppleWorks, you have to use slashes.

The IIGS Desktop is supposed to protect us from having to concern ourselves about directories by representing them graphically. We mention the concept for two reasons:

First, a lot of older programs still use the subdirectory terminology, so it's important to relate the two ways of talking about your disk contents. Graphic user interfaces are relatively new, and the Apple II's been around for a *long* time.

Second, a lot of old-time Apple users are familiar with the subdirectory concept but have less experience with windows and icons. The two concepts are just different ways of thinking about a hierarchical organization, and that's important to know.

Copying & Moving Files

To copy or move files, just select the files and drag them from one window to another (or to a disk or folder icon). If you drag the icons to a different disk, the files will be copied to the target disk (the original copies will remain intact on the source disk). If you drag the icons to a different folder on the same disk (or to the disk's main directory) the files will be moved to the new location.

If you drag a folder, everything in the folder (including any folders and their contents, even folders inside folders inside folders) will be copied or moved.

Here's an example. Let's assume that we've received some fonts we want to install on our hard drive. The Fonts folder is inside the System folder on our Q Drive; the name of the Q Drive's first partition is Q1. The new fonts themselves are in the main directory of a 3.5" disk and are the only files on the disk.

We'll start with all windows closed. Next, we'll open our Q Drive's Q1 icon (double-click) and tunnel into the System folder (hold down Option while double-clicking the System folder). We'll locate the Fonts folder by scrolling the window until it is visible.

Now we'll open the 3.5" disk containing the fonts we want to install. We may need to position our windows so they don't overlap too much—we need to be able to see the Fonts folder in the System window so we can drag the new fonts into it. Next we'll bring the window containing our new fonts to the front by clicking in it once, then select all files by pressing ⌘-A. Finally, we'll drag the font files to the Fonts folder on our Q Drive, making sure that the pointer is directly on the Fonts folder and that the Fonts folder is blackened before releasing the mouse button. (If the Fonts folder isn't black, the Finder will copy the fonts into the System folder.) We release the mouse button and copying begins.

If some of the fonts we're copying are already in the Fonts folder on our hard drive, the Finder will give us the choice of replacing them or skipping over duplicate items. We'll probably click "Skip" under All, since we'd just be wasting time copying those fonts.

➡ **Shortcut**

To specify in advance how you want the Finder to handle items that already exist in the target directory, hold down the Option key as you drag the files. The Alternatives dialog has buttons that allow you to resolve duplicates on a case-by-case basis or to replace all or no duplicates.

Yes, it really is that simple. We could have opened the Fonts folder and dragged the new fonts into the Fonts window—remember, the window and the folder it is linked to are, essentially, the same thing. If the files we were copying were in a folder on the 3.5" disk, we would have tunneled into that folder after opening the disk—the rest of the procedure would be the same.

Here's another example, this time focusing on moving (not copying) files. Let's assume we've got some AppleWorks documents in our hard drive's main directory that belong in a folder. The folder is called AW.Files and is inside the AppleWorks folder, which is in the hard drive's main directory, Q1. As before, we'll start with all windows closed.